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A nonconvex ADMM for a class of sparse inverse semidefinite quadratic programming problems

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ABSTRACT

In this paper, we consider a class of sparse inverse semidefinite quadratic programming problems, in which a nonconvex alternating direction method of multiplier is investigated. Under mild conditions, we establish convergence results of our algorithm and the corresponding non-ergodic iteration-complexity is also considered under the assumption that the potential function satisfies the famous Kurdyka–Łojasiewicz property. Numerical results show that our algorithm is suitable to solve the given sparse inverse semidefinite quadratic programming problems.

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1. Introduction

In the usual optimization process, all parameters appeared in the model are given, and the main task is to find an optimal solution for a specified objective function with constraints. More specifically, consider an optimization problem of the form

$$\begin{aligned} \min \quad & f(s; \theta) \\ \text{s.t.} \quad & s \in \mathcal{M}, \end{aligned} \tag{1}$$

where θ is a parameter and s is the decision variable. However, in some applications, the situation is reversed and we focus on finding values of parameters in Problem (1), which also make a given feasible solution optimal and the parameters differ from their estimates (obtained from experience and observation) as

little as possible. In this case, these problems are called inverse optimization problems, which look as follows

$$\begin{aligned} \min \quad & \|\theta - \theta^e\| \\ \text{s.t.} \quad & \theta \in \Theta(s^e), \end{aligned}$$

where θ^e is a nominal parameter value and $\|\cdot\|$ is an appropriate norm. For a given feasible solution s^e , $\Theta(s^e)$ denotes the collection of θ for which s^e is optimal to (1).

The systematic study of this issue began in some shortest paths problems initially analysed by Burton and Toint [1], whose results triggered off a series of contributions on inverse network problems [2–7]. After that, several special inverse continuous optimization models appear in the literatures. For instances, inverse linear programming problem [8,9], inverse second-order cone programming problem [10] and inverse quadratic programming problem [11]. In the past decade, matrix optimization problems attract much attention, particularly in semidefinite programming (SDP). Consequently, plenty of numerical algorithms (e.g., sequential convex approximation, semismooth Newton method and alternating direction method of multipliers) were developed for a class of inverse semidefinite linear programming problem [12] and inverse semidefinite quadratic programming problem [13–15]. However, these models in some sense do not take into account the data feature of parameters, which is key to numerical algorithm designs.

We introduce some notations used in the sequel. Let $\mathcal{S}^n$ be the space of all real-valued $n \times n$ symmetric matrices endowed with the trace inner product $\langle \cdot, \cdot \rangle$ and the Frobenius norm $\|\cdot\|_F$. The cone $\mathcal{S}_+^n$ consists of all real-valued $n \times n$ positive semidefinite matrices and $C \succeq D$ if and only if $C - D \in \mathcal{S}_+^n$. Similarly, we denote by $\mathcal{S}_{++}^n$ the set of all real-valued positive definite matrices of order n and $C \succ D$ means that $C - D \in \mathcal{S}_{++}^n$. We use $\langle \cdot, \cdot \rangle_v$ and $\|\cdot\|_2$ to denote the inner product and the Euclidean norm defined on $\mathbb{R}^n$, respectively. Given a family of matrices $\{A_1, A_2, \dots, A_n\}$ in $\mathcal{S}^m$, the operator $\mathcal{A} : \mathbb{R}^n \rightarrow \mathcal{S}^m$ and its adjoint operator $\mathcal{A}^* : \mathcal{S}^m \rightarrow \mathbb{R}^n$ are respectively described as $\mathcal{A}(x) := \sum_{i=1}^n x_i A_i \in \mathcal{S}^m$ and $\mathcal{A}^*(X) := [\langle A_1, X \rangle, \langle A_2, X \rangle, \dots, \langle A_n, X \rangle]^T \in \mathbb{R}^n$.

1.1. Motivation

We take an interesting portfolio optimization problem to show the necessity of sparse inverse semidefinite quadratic problem (see Section 1.2 for more details) in practice.

Let us consider the following situation: An investor expects to find the best plan of investing on n stocks in a stock market. Let μ_i denote the expected return of i th stock. The covariance matrix among these stocks return is denoted by $C \in \mathbb{R}^{n \times n}$, it is reasonable to assume that the matrix $C = (c_{ij})$ must be at least positive

semidefinite, where c_{ij} is the covariance between returns of stocks i and j . The portfolio optimization problem can be described as

$$\begin{aligned} \min \quad & \frac{1}{2}x^T Cx + \gamma \mu^T x \\ \text{s.t.} \quad & B - \mathcal{A}(x) \succeq 0, \\ & x \in \mathbb{R}^n, \end{aligned} \tag{2}$$

where $x := (x_1, x_2, \dots, x_n)^T \in \mathbb{R}^n$ denotes the investment proportion vector and x_i is the investment proportion on stock i . $\mu := (\mu_1, \mu_2, \dots, \mu_n)^T \in \mathbb{R}^n$ denotes the expected return vector of stocks. $\gamma < 0$ is a trade-off parameter, which uses to control the balance between return and risk. The constraint $B - \mathcal{A}(x) \succeq 0$ describes a limitation of investment by choosing parameters A_i ($i = 1, 2, \dots, n$) and B properly. It is easy to see that the above problem can be viewed as a class of semidefinite quadratic programming problem.

Let $(\bar{C}, \bar{\mu})$ be a current estimate of (C, μ) and x^e be the optimal solution to (2) with $(C, \mu) = (\bar{C}, \bar{\mu})$. Suppose that the investor now acquires updated information on these target stocks, e.g., the new estimate (C^e, μ^e) . The investor will be faced with a difficult choice about whether the investment proportion vector x^e is required to be modified, because the corresponding adjustment may result in a loss or get a high return. In addition, for a short-term investment, it is reasonable to assume that the difference between C and C^e is sparse. In order to deal with the above issue, we consider the following inverse optimization problem

$$\begin{aligned} \min \quad & f(C, \mu) := \|C - C^e\|_1 + \|\mu - \mu^e\|_2^2 \\ \text{s.t.} \quad & (C, \mu) \in \Theta(x^e), \\ & (C, \mu) \in \mathcal{S}_+^n \times \mathbb{R}^n, \end{aligned}$$

where $\Theta(x^e)$ denotes the collection of (C, μ) for which x^e is optimal to (2). If $(C^e, \mu^e) \in \Theta(x^e)$, then $f(C^e, \mu^e) = 0$, x^e is also an optimal solution to (2) with $(C, \mu) = (C^e, \mu^e)$, therefore the investor does not need to adjust the proportion x^e . Otherwise, $f(C^e, \mu^e) > 0$, the above inverse optimization model will balance the risk and return automatically. Let f^t be the investor's threshold on adjustment, if the optimal objective function value f^* is greater than f^t , the investor needs to revise his proportion vector, otherwise unchanged. In any case, it is necessary to study the above inverse semidefinite quadratic programming problem.

1.2. The problem description

In this paper, we consider the following sparse inverse semidefinite quadratic programming problem (S-ISDQP(G, c))

$$\begin{aligned} \min \quad & \Phi(G - G^e) + \frac{1}{2} \|c - c^e\|_2^2 \\ \text{s.t.} \quad & x^e \in \text{Sol}(\text{SDQP}(G, c)), \\ & (G, c) \in \mathcal{S}_+^n \times \mathbb{R}^n, \end{aligned} \quad (3)$$

where Φ is a proper, closed, nonnegative, possibly nonconvex function defined on $\mathbb{R}^{n \times n}$ and has a generic form as $\Phi(Y) := \sum_{i=1}^n \sum_{j=1}^n \phi(Y_{ij})$ with a lower semi-continuous nonnegative function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ such that $\phi(0) = 0$. We list some examples of $\phi(t)$, $t \in \mathbb{R}$ as follows:

- The l_1 regularizer: $\phi(t) := |t|$;
- The l_2 regularizer: $\phi(t) := |t|^2$;
- The l_q regularizer: $\phi(t) := |t|^q$, $q \in (0, 1)$.

Here $\text{Sol}(\cdot)$ denotes the optimal solution mapping of a given problem, $(G^e, c^e) \in \mathbb{R}^{n \times n} \times \mathbb{R}^n$ is an estimate to (G, c) and $x^e \in \mathbb{R}^n$ is a given feasible point of semidefinite quadratic programming problem (SDQP(G, c))

$$\begin{aligned} \min \quad & \frac{1}{2} x^T G x + c^T x \\ \text{s.t.} \quad & B - \mathcal{A}(x) \succeq 0, \\ & x \in \mathbb{R}^n, \end{aligned}$$

where $B \in \mathcal{S}^m$ and $G \in \mathcal{S}_+^n$. It is worth noting that if we set $\phi(t) := \frac{1}{2}|t|^2$, then Problem S-ISDQP(G, c) reduces to the ordinary inverse semidefinite quadratic problem [13–15]. Therefore, Problem S-ISDQP(G, c) is a generalization of ISDQP(G, c) and the term $\Phi(G - G^e)$ characterizes the above sparsity assumption on the variable G .

Assume that Problem SDQP(G, c) has an attainable optimal value and the Slater's condition holds (i.e., there exists a vector $\bar{x} \in \mathbb{R}^n$ satisfying $B - \mathcal{A}(\bar{x}) \succ 0$), then $x^e \in \text{Sol}(\text{SDQP}(G, c))$ if and only if there exists a matrix $\Omega \in \mathcal{S}^m$ such that

$$\begin{cases} c + Gx^e - \mathcal{A}^*(\Omega) = 0, \\ \Omega \in \mathcal{S}_+^m, Z^e \in \mathcal{S}_+^m, \langle \Omega, Z^e \rangle = 0, \end{cases}$$

where $Z^e := B - \mathcal{A}(x^e) \in \mathcal{S}^m$. In addition, if the rank of Z^e is equal to r , according to the eigenvalue decomposition, there exist an orthogonal matrix $Q \in \mathbb{R}^{m \times m}$

and a diagonal matrix $\Lambda_r \in \mathcal{S}^r$ such that

$$Z^e = [Q_r \ Q_{r^c}] \begin{bmatrix} \Lambda_r & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} Q_r^T \\ Q_{r^c}^T \end{bmatrix}$$

with $Q := [Q_r \ Q_{r^c}] \in \mathbb{R}^{m \times m}$, $Q_r \in \mathbb{R}^{m \times r}$, $Q_{r^c} \in \mathbb{R}^{m \times p}$, $p := m - r$ and $\Lambda_r := \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_r)$, $\lambda_i \in \mathbb{R}$ is the i -th positive eigenvalue of Z^e . Under the above settings, we further define $\hat{A}_i := Q_{r^c}^T A_i Q_{r^c} \in \mathcal{S}^p$, $\hat{\Omega} := Q_{r^c}^T \Omega Q_{r^c} \in \mathcal{S}^p$, Problem S-ISDQP(G, c) now can be described as

$$\begin{aligned} \min_{G \in \mathcal{S}^n, \hat{\Omega} \in \mathcal{S}^p} \quad & F(G, \hat{\Omega}) := \Phi(G - G^e) + \delta_{\mathcal{S}_+^n}(G) \\ & + \frac{1}{2} \|\hat{A}^*(\hat{\Omega}) - Gx^e - c^e\|_2^2 + \delta_{\mathcal{S}_+^p}(\hat{\Omega}), \end{aligned} \quad (4)$$

where $\delta_{\mathcal{S}_+^n}(\cdot)$ is the indicator function of $\mathcal{S}_+^n$. Notice that the objective function $F(G, \hat{\Omega})$ of (4) may now become a nonconvex function due to different choices of ϕ .

Recently, Łojasiewicz [16,17] proposes a unified approach for solving a class of nonconvex optimization problems, in which their potential functions satisfy the famous Kurdyka–Łojasiewicz (KL) property (see Definition 2.2 below). Besides these results, some variants of classical optimization algorithms (e.g., proximal point algorithm [18–21] and splitting methods [22]) are also studied under such assumption, which motivate us to design a numerical algorithm for Problem S-ISDQP(G, c) given as in (4).

1.3. Contribution and organization

In this paper, we propose a nonconvex alternating direction method of multipliers (ADMM) for a class of sparse inverse semidefinite quadratic programming problems, in which it combines a nontrivial dual stepsize (potentially larger than (1) with the classical ADMM [15]. Our contribution is to establish a non-ergodic iteration-complexity bound for Problem S-ISDQP(G, c) under a suitable assumption on the KL property, while there are no such discussions in related literatures [14,15,23–26]. It is well known that complexity analysis is an indispensable ingredient in modern optimization, see [27] and references therein. At the same time, a computable thresholding of penalty parameter and a first iterative condition are also studied, which are used to analyse convergence results of our algorithm.

The remainder of this paper is organized as follows. In Section 2, we first recall some preliminary results on the KL property. In Section 3, we present a nonconvex ADMM for Problem S-ISDQP(G, c), in which optimality conditions of its subproblems are analysed and convergence results of our method are also established. Moreover, we discuss the corresponding non-ergodic

iteration-complexity under the KL property. Numerical results are presented in Section 4. Finally, we give some concluding remarks.

2. Preliminaries

In this section, we first recall some background materials about subdifferentials in variational analysis [28] and then present some technical results on the KL property.

Definition 2.1: Let $f : \mathcal{S}^n \rightarrow (-\infty, +\infty]$ be a proper closed function and its effective domain $\text{dom} f := \{X \in \mathcal{S}^n : f(X) < +\infty\}$. The limiting subdifferential (or simply the subdifferential) of f at $X \in \text{dom} f$, denoted by $\partial f(X)$, is defined as

$$\partial f(X) := \left\{ D \in \mathcal{S}^n : \exists X^k \rightarrow X, f(X^k) \rightarrow f(X), D^k \rightarrow D \text{ with } D^k \in \hat{\partial} f(X^k) \right\},$$

where $\hat{\partial} f(X)$ stands for the Fréchet subdifferential of f at X , which is given by

$$\hat{\partial} f(X) := \left\{ D \in \mathcal{S}^n : \liminf_{Y \rightarrow X, Y \neq X} \frac{f(Y) - f(X) - \langle D, Y - X \rangle}{\|Y - X\|_F} \geq 0 \right\}.$$

For any $X \in \mathcal{S}_+^n$, the normal cone of $\mathcal{S}_+^n$ at X is expressed as

$$\mathcal{N}_{\mathcal{S}_+^n}(X) := \{Y \in \mathcal{S}^n : \langle Y, X' - X \rangle \leq 0, \forall X' \in \mathcal{S}_+^n\}.$$

From Definition 2.1, we have $\mathcal{N}_{\mathcal{S}_+^n}(X) = \partial \delta_{\mathcal{S}_+^n}(X)$ for any $X \in \mathcal{S}_+^n$. Additionally, for a proper function $g(X, Y) : \mathcal{S}^n \times \mathcal{S}^n \rightarrow (-\infty, +\infty]$, we write $\partial_X g(\bar{X}, \bar{Y})$ the partial subdifferential of g at $(\bar{X}, \bar{Y})$ with respect to the variable X .

Let $(G^*, \hat{\Omega}^*)$ be a local minimizer of Problem S-ISDQP(G, c), the generalized Fermat's rule [28, Theorem 10.1] yields that

$$0 \in \partial F(G^*, \hat{\Omega}^*) := \left[\begin{array}{c} \partial \Phi(G^* - G^e) + \mathcal{L}_{x^e}(c^e + G^* x^e - \hat{A}^*(\hat{\Omega}^*)) + \mathcal{N}_{\mathcal{S}_+^n}(G^*) \\ \hat{A}(\hat{A}^*(\hat{\Omega}^*) - G^* x^e - c^e) + \mathcal{N}_{\mathcal{S}_+^p}(\hat{\Omega}^*) \end{array} \right], \quad (5)$$

where F is defined as in (4) and $\mathcal{L}_{x^e}(c^e + G^* x^e - \hat{A}^*(\hat{\Omega}^*))$ is given by

$$\begin{aligned} & \mathcal{L}_{x^e}(c^e + G^* x^e - \hat{A}^*(\hat{\Omega}^*)) \\ &:= \frac{x^e(c^e + G^* x^e - \hat{A}^*(\hat{\Omega}^*))^T + (c^e + G^* x^e - \hat{A}^*(\hat{\Omega}^*))(x^e)^T}{2}. \end{aligned} \quad (6)$$

In addition, we denote by $S(G, \hat{\Omega})$ the set of stationary points for Problem, i.e., $S(G, \hat{\Omega}) := \{(G, \hat{\Omega}) : 0 \in \partial F(G, \hat{\Omega})\}$, where ∂F is defined as in (5).

To end this section, we recall the definition of the KL property [21, Definition 3]. In what follows, we denote by Γ_ω ($\omega \in (0, +\infty]$) a class of concave functions $\sigma : [0, \omega) \rightarrow \mathbb{R}_+$ satisfying the following three conditions:

- (a) σ is continuous differentiable on $(0, \omega)$;
- (b) $\lim_{s \rightarrow 0} \sigma(s) = \sigma(0) = 0$;
- (c) For any $s \in (0, \omega)$, the derivative $\sigma'(s)$ is strictly positive.

Definition 2.2: Let $f : \mathcal{S}^n \rightarrow (-\infty, +\infty]$ be a proper, lower semicontinuous function, we say f to have the KL property at $\bar{X} \in \text{dom} \partial f := \{X \in \mathcal{S}^n : \partial f(X) \neq \emptyset\}$, if there exist $\omega \in (0, +\infty]$, a neighbourhood V of $\bar{X}$ and a function $\sigma \in \Gamma_\omega$ such that for all $X \in V \cap \{X \in \mathcal{S}^n : 0 < f(X) - f(\bar{X}) < \omega\}$, the inequality $\sigma'(f(X) - f(\bar{X})) \inf_{Y \in \partial f(X)} \|Y\|_F \geq 1$ holds at $\bar{X}$. If f satisfies the KL property at each point of $\text{dom} \partial f$, then f is called a KL function.

An important result on the KL property is described in the uniform sense. We refer the interested readers to [21, Lemma 6] for the proof.

Lemma 2.3: Let Ξ be a compact set and $f : \mathcal{S}^n \rightarrow (-\infty, +\infty]$ be a proper, lower semicontinuous function. Suppose that f is constant on Ξ and satisfies the KL property at each point of Ξ . Then there exist $\epsilon > 0, \omega > 0$ and $\sigma \in \Gamma_\omega$ such that $\sigma'(f(X) - f(\bar{X})) \inf_{Y \in \partial f(X)} \|Y\|_F \geq 1$ for all $\bar{X} \in \Xi$ and $X \in \{X \in \mathcal{S}^n : \inf_{Y \in \Xi} \|X - Y\|_F \leq \epsilon\} \cap \{X \in \mathcal{S}^n : 0 < f(X) - f(\bar{X}) < \omega\}$.

3. A nonconvex ADMM for problem S-ISDQP(G, c)

In this section, we present a nonconvex ADMM for Problem S-ISDQP(G, c). To begin, through importing an auxiliary variable $W \in \mathcal{S}^n$ as the way in [16,26], Problem (4) reduces to the following form

$$\begin{aligned}
 \min \quad & P(G, \hat{\Omega}, W) := \Phi(G - G^e) + \delta_{\mathcal{S}_+^n}(G) \\
 & + \frac{1}{2} \|\hat{\mathcal{A}}^*(\hat{\Omega}) - Wx^e - c^e\|_2^2 + \delta_{\mathcal{S}_+^p}(\hat{\Omega}), \\
 \text{s.t.} \quad & W - G = 0, \\
 & (G, \hat{\Omega}, W) \in \mathcal{S}^n \times \mathcal{S}^p \times \mathcal{S}^n.
 \end{aligned} \tag{7}$$

We next introduce the associated augmented Lagrangian function

$$l_\beta(G, \hat{\Omega}, W, \Lambda) := P(G, \hat{\Omega}, W) + \langle \Lambda, W - G \rangle + \frac{\beta}{2} \|W - G\|_F^2, \tag{8}$$

where $P(G, \hat{\Omega}, W)$ is defined as in (7), $\Lambda \in \mathcal{S}^n$ denotes the multiplier of the equality constraint in (7) and $\beta > 0$ is the penalty parameter. The details of our method is displayed in Algorithm 1.

Algorithm 1 A nonconvex ADMM for Problem S-ISDQP(G, c)

Input: $(\hat{\Omega}^0, W^0, \Lambda^0) \in \mathcal{S}_+^p \times \mathcal{S}^n \times \mathcal{S}^n$, $\tau \in (0, \frac{1+\sqrt{5}}{2})$, $\beta > 0$ and $k := 0$.
while the termination criterion is not met **do**
 1: Solve the G -subproblem: $G^{k+1} \in \text{Argmin}_{G \in \mathcal{S}^n} l_\beta(G, \hat{\Omega}^k, W^k, \Lambda^k)$.
 2: Solve the $\hat{\Omega}$ -subproblem: $\hat{\Omega}^{k+1} \in \text{Argmin}_{\hat{\Omega} \in \mathcal{S}^p} l_\beta(G^{k+1}, \hat{\Omega}, W^k, \Lambda^k)$.
 3: Solve the W -subproblem: $W^{k+1} \in \text{Argmin}_{W \in \mathcal{S}^n} l_\beta(G^{k+1}, \hat{\Omega}^{k+1}, W, \Lambda^k)$.
 4: Update the multiplier and the counter: $\Lambda^{k+1} := \Lambda^k + \tau\beta(W^{k+1} - G^{k+1})$,
 $k := k + 1$.
end while
Output: $(G^k, \hat{\Omega}^k, W^k, \Lambda^k)$.

3.1. Convergence analysis

In this subsection, we present convergence results of Algorithm 1. We first analyse optimality conditions of three subproblems. For the G -subproblem, the generalized Fermat's rule [28, Theorem 10.1] implies that

$$0 \in \partial\Phi(G^{k+1} - G^e) + \mathcal{N}_{\mathcal{S}_+^p}(G^{k+1}) - \Lambda^k + \beta(G^{k+1} - W^k). \quad (9)$$

Similarly, the first-order necessary conditions of the $\hat{\Omega}$ -subproblem and the W -subproblem are respectively given by

$$0 \in \hat{\mathcal{A}}(\hat{\mathcal{A}}^*(\hat{\Omega}^{k+1}) - W^k x^e - c^e) + \mathcal{N}_{\mathcal{S}_+^p}(\hat{\Omega}^{k+1}), \quad (10)$$

$$\mathcal{L}_{x^e}(c^e + W^{k+1} x^e - \hat{\mathcal{A}}^*(\hat{\Omega}^{k+1})) + \Lambda^k + \beta(W^{k+1} - G^{k+1}) = 0. \quad (11)$$

Our convergence analysis relies on the following assumption.

Assumption 3.1: There exists a positive constant $\theta_{\hat{\mathcal{A}}}$ such that $\langle \hat{\Omega}, \hat{\mathcal{A}}\hat{\mathcal{A}}^*(\hat{\Omega}) \rangle \geq \theta_{\hat{\mathcal{A}}}^2 \|\hat{\Omega}\|_F^2$, $\forall \hat{\Omega} \in \mathcal{S}^p$.

Remark 3.1: The assumption on $\hat{\mathcal{A}}$ comes from [15, Section 3.2], which is also commonly used in nonconvex ADMMs, see [25, Assumption 3.1] or [26, Assumption 4.1(a)].

In the sequel, we denote by $\Psi_{\tau, \beta}(G, \hat{\Omega}, W, \Lambda)$ the potential function of Problem S-ISDQP(G, c) as follows:

$$\Psi_{\tau, \beta}(G, \hat{\Omega}, W, \Lambda) := l_\beta(G, \hat{\Omega}, W, \Lambda) + \max \left\{ 1 - \tau, \frac{(\tau - 1)\tau^2}{1 + \tau - \tau^2} \right\} \beta \|W - G\|_F^2. \quad (12)$$

Remark 3.2: This potential function has two parts, the classical augmented Lagrangian function and the extra penalty term for the equality constraint in (7).

Moreover, for any $\tau \in (0, (1 + \sqrt{5})/2)$, the coefficient of the extra penalty term is nonnegative and hence the potential function $\Psi_{\tau,\beta}(G, \hat{\Omega}, W, \Lambda)$ majorizes the augmented Lagrangian function $l_\beta(G, \hat{\Omega}, W, \Lambda)$ and $\Psi_{1,\beta}(G, \hat{\Omega}, W, \Lambda)$ equals to $l_\beta(G, \hat{\Omega}, W, \Lambda)$.

We start our analysis on the difference of $\Psi_{\tau,\beta}$ at two successive iteration points.

Lemma 3.1: Suppose that $\tau \in (0, (1 + \sqrt{5})/2)$ and $\{(G^k, \hat{\Omega}^k, W^k, \Lambda^k)\}_{k=1}^\infty$ is a sequence generated by Algorithm 1. If Assumption 3.1 holds, then there exist two positive constants $\theta_{\hat{\Omega}}, \theta_W$ such that

$$\begin{aligned} & \Psi_{\tau,\beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k) - \Psi_{\tau,\beta}(G^{k+1}, \hat{\Omega}^{k+1}, W^{k+1}, \Lambda^{k+1}) \\ & \geq \left[\frac{\theta_{\hat{\Omega}}^2}{2} - \max \left\{ \frac{1}{\tau}, \frac{\tau^2}{1 + \tau - \tau^2} \right\} \frac{\theta_{\hat{\Omega}}^2}{\beta} \right] \|\hat{\Omega}^{k+1} - \hat{\Omega}^k\|_F^2 \\ & \quad + \left[\frac{\beta}{2} - \max \left\{ \frac{1}{\tau}, \frac{\tau^2}{1 + \tau - \tau^2} \right\} \frac{\theta_W^2}{\beta} \right] \|W^{k+1} - W^k\|_F^2. \end{aligned} \quad (13)$$

Proof: From the definition of $\Psi_{\tau,\beta}(G, \hat{\Omega}, W, \Lambda)$ given as in (12), we have

$$\begin{aligned} & \Psi_{\tau,\beta}(G^{k+1}, \hat{\Omega}^{k+1}, W^{k+1}, \Lambda^{k+1}) - \Psi_{\tau,\beta}(G^{k+1}, \hat{\Omega}^{k+1}, W^{k+1}, \Lambda^k) \\ & = l_\beta(G^{k+1}, \hat{\Omega}^{k+1}, W^{k+1}, \Lambda^{k+1}) - l_\beta(G^{k+1}, \hat{\Omega}^{k+1}, W^{k+1}, \Lambda^k) \\ & = \langle \Lambda^{k+1} - \Lambda^k, \frac{1}{\tau\beta}(\Lambda^{k+1} - \Lambda^k) \rangle \\ & = \frac{1}{\tau\beta} \|\Lambda^{k+1} - \Lambda^k\|_F^2, \end{aligned} \quad (14)$$

where the second equation follows from Step 4 of Algorithm 1. Moreover, Equation (11) is equivalent to

$$\begin{aligned} \Lambda^{k+1} - \Lambda^k & = \tau \mathcal{L}_{x^e}(\hat{\mathcal{A}}^*(\hat{\Omega}^{k+1} - \hat{\Omega}^k) - (W^{k+1} - W^k)x^e) \\ & \quad + (1 - \tau)(\Lambda^k - \Lambda^{k-1}). \end{aligned} \quad (15)$$

From the definition of $\hat{\mathcal{A}}$, we have

$$\|\mathcal{L}_{x^e}(\hat{\mathcal{A}}^*(\hat{\Omega}^{k+1} - \hat{\Omega}^k))\|_F^2 \leq \theta_{\hat{\Omega}}^2 \|\hat{\Omega}^{k+1} - \hat{\Omega}^k\|_F^2, \quad (16)$$

$$\|\mathcal{L}_{x^e}((W^{k+1} - W^k)x^e)\|_F^2 \leq \theta_W^2 \|W^{k+1} - W^k\|_F^2, \quad (17)$$

where $\theta_{\hat{\Omega}} := n\sqrt{\sum_{i=1}^n \sum_{j=1}^n \|x_i^e \hat{\mathcal{A}}_j\|_\infty^2}$ and $\theta_W := \sqrt{n\|x^e\|_\infty^2 \|x^e\|_2^2}$. According to Equations (15)–(17), we discuss two cases:

Case 1: If $\tau \in (0, 1]$, due to the convexity of $\|\cdot\|_F^2$, we have

$$\begin{aligned} & \|\Lambda^{k+1} - \Lambda^k\|_F^2 \\ & \leq \tau \left(\|\mathcal{L}_{x^e}(\hat{\mathcal{A}}^*(\hat{\Omega}^{k+1} - \hat{\Omega}^k))\|_F^2 + \|\mathcal{L}_{x^e}((W^{k+1} - W^k)x^e)\|_F^2 \right) \\ & \quad + (1 - \tau) \|\Lambda^k - \Lambda^{k-1}\|_F^2 \\ & \leq \tau \left(\theta_{\hat{\Omega}}^2 \|\hat{\Omega}^{k+1} - \hat{\Omega}^k\|_F^2 + \theta_W^2 \|W^{k+1} - W^k\|_F^2 \right) + (1 - \tau) \|\Lambda^k - \Lambda^{k-1}\|_F^2. \end{aligned}$$

Case 2: If $\tau \in (1, (1 + \sqrt{5})/2)$, similar to Case 1, we have

$$\begin{aligned} & \left\| \frac{1}{\tau} (\Lambda^{k+1} - \Lambda^k) \right\|_F^2 \\ & \leq \frac{1}{\tau} \left(\tau^2 \|\mathcal{L}_{x^e}(\hat{\mathcal{A}}^*(\hat{\Omega}^{k+1} - \hat{\Omega}^k))\|_F^2 + \tau^2 \|\mathcal{L}_{x^e}((W^{k+1} - W^k)x^e)\|_F^2 \right) \\ & \quad + \left(1 - \frac{1}{\tau} \right) \|\Lambda^k - \Lambda^{k-1}\|_F^2, \end{aligned}$$

which implies that $\|\Lambda^{k+1} - \Lambda^k\|_F^2 \leq \tau^3 (\theta_{\hat{\Omega}}^2 \|\hat{\Omega}^{k+1} - \hat{\Omega}^k\|_F^2 + \theta_W^2 \|W^{k+1} - W^k\|_F^2) + (\tau^2 - \tau) \|\Lambda^k - \Lambda^{k-1}\|_F^2$. It follows from the above results in Case 1 and Case 2 that

$$\begin{aligned} & \frac{1}{\tau\beta} \|\Lambda^{k+1} - \Lambda^k\|_F^2 \\ & \leq \max \left\{ \frac{1}{\tau}, \frac{\tau^2}{1 + \tau - \tau^2} \right\} \left(\frac{\theta_{\hat{\Omega}}^2}{\beta} \|\hat{\Omega}^{k+1} - \hat{\Omega}^k\|_F^2 + \frac{\theta_W^2}{\beta} \|W^{k+1} - W^k\|_F^2 \right) \\ & \quad + \max \left\{ 1 - \tau, \frac{(\tau^2 - \tau)\tau}{1 + \tau - \tau^2} \right\} \frac{1}{\tau^2\beta} \left(\|\Lambda^k - \Lambda^{k-1}\|_F^2 - \|\Lambda^{k+1} - \Lambda^k\|_F^2 \right). \end{aligned} \quad (18)$$

The inequality (18) together with Equation (14) imply that

$$\begin{aligned} & \Psi_{\tau,\beta}(G^{k+1}, \hat{\Omega}^{k+1}, W^{k+1}, \Lambda^{k+1}) - \Psi_{\tau,\beta}(G^{k+1}, \hat{\Omega}^{k+1}, W^{k+1}, \Lambda^k) \\ & \leq \max \left\{ \frac{1}{\tau}, \frac{\tau^2}{1 + \tau - \tau^2} \right\} \left(\frac{\theta_{\hat{\Omega}}^2}{\beta} \|\hat{\Omega}^{k+1} - \hat{\Omega}^k\|_F^2 + \frac{\theta_W^2}{\beta} \|W^{k+1} - W^k\|_F^2 \right) \\ & \quad + \max \left\{ 1 - \tau, \frac{(\tau^2 - \tau)\tau}{1 + \tau - \tau^2} \right\} \beta \left(\|W^k - G^k\|_F^2 - \|W^{k+1} - G^{k+1}\|_F^2 \right). \end{aligned} \quad (19)$$

Similar to the first part, we also obtain

$$\begin{aligned}
& \Psi_{\tau,\beta}(G^{k+1}, \hat{\Omega}^{k+1}, W^{k+1}, \Lambda^k) - \Psi_{\tau,\beta}(G^{k+1}, \hat{\Omega}^{k+1}, W^k, \Lambda^k) \\
&= l_\beta(G^{k+1}, \hat{\Omega}^{k+1}, W^{k+1}, \Lambda^k) - l_\beta(G^{k+1}, \hat{\Omega}^{k+1}, W^k, \Lambda^k) \\
&\quad + \max \left\{ 1 - \tau, \frac{(\tau - 1)\tau^2}{1 + \tau - \tau^2} \right\} \beta \left(\|W^{k+1} - G^{k+1}\|_F^2 - \|W^k - G^{k+1}\|_F^2 \right) \\
&= \frac{1}{2} \|(\hat{A}^*(\hat{\Omega}^{k+1}) - W^k x^e - c^e) - (W^{k+1} - W^k) x^e\|_2^2 \\
&\quad - \frac{1}{2} \|\hat{A}^*(\hat{\Omega}^{k+1}) - W^k x^e - c^e\|_2^2 \\
&\quad + \langle \Lambda^k, W^{k+1} - W^k \rangle + \frac{\beta}{2} \left(\|W^{k+1} - G^{k+1}\|_F^2 - \|W^k - G^{k+1}\|_F^2 \right) \\
&\quad + \max \left\{ 1 - \tau, \frac{(\tau - 1)\tau^2}{1 + \tau - \tau^2} \right\} \beta \left(\|W^{k+1} - G^{k+1}\|_F^2 - \|W^k - G^{k+1}\|_F^2 \right) \\
&= \left\langle \mathcal{L}_{x^e} \left(c^e + W^{k+1} x^e - \hat{A}^*(\hat{\Omega}^{k+1}) \right) + \Lambda^k, W^{k+1} - W^k \right\rangle \\
&\quad - \frac{\beta}{2} \left(\|W^k - W^{k+1}\|_F^2 + 2 \left\langle W^{k+1} - G^{k+1}, W^k - W^{k+1} \right\rangle \right) \\
&\quad + \max \left\{ 1 - \tau, \frac{(\tau - 1)\tau^2}{1 + \tau - \tau^2} \right\} \beta \left(\|W^{k+1} - G^{k+1}\|_F^2 - \|W^k - G^{k+1}\|_F^2 \right) \\
&= \max \left\{ 1 - \tau, \frac{(\tau - 1)\tau^2}{1 + \tau - \tau^2} \right\} \beta \left(\|W^{k+1} - G^{k+1}\|_F^2 - \|W^k - G^{k+1}\|_F^2 \right) \\
&\quad - \frac{\beta}{2} \|W^k - W^{k+1}\|_F^2, \tag{20}
\end{aligned}$$

where the last equation is obtained by Equation (11). On the other hand, Assumption 3.1 and Equation (10) show that

$$\begin{aligned}
& \Psi_{\tau,\beta}(G^{k+1}, \hat{\Omega}^{k+1}, W^k, \Lambda^k) - \Psi_{\tau,\beta}(G^{k+1}, \hat{\Omega}^k, W^k, \Lambda^k) \\
&= l_\beta(G^{k+1}, \hat{\Omega}^{k+1}, W^k, \Lambda^k) - l_\beta(G^{k+1}, \hat{\Omega}^k, W^k, \Lambda^k) \\
&\leq - \left\langle \hat{A} \left(\hat{A}^*(\hat{\Omega}^{k+1}) - W^k x^e - c^e \right) + \eta^{k+1}, \hat{\Omega}^k - \hat{\Omega}^{k+1} \right\rangle \\
&\quad - \frac{1}{2} \|\hat{A}^*(\hat{\Omega}^k - \hat{\Omega}^{k+1})\|_2^2 \\
&= -\frac{\theta^2}{2} \|\hat{\Omega}^k - \hat{\Omega}^{k+1}\|_F^2, \tag{21}
\end{aligned}$$

where η^{k+1} is an element of $\mathcal{N}_{\mathcal{S}_+^p}(\hat{\Omega}^{k+1})$ such that $\hat{A}(\hat{A}^*(\hat{\Omega}^{k+1}) - W^k x^e - c^e) + \eta^{k+1} = 0$. Since G^{k+1} is an optimal solution of the G -subproblem

$\min_{G \in \mathcal{S}^n} l_\beta(G, \hat{\Omega}^k, W^k, \Lambda^k)$, we have

$$\begin{aligned} & \Psi_{\tau, \beta}(G^{k+1}, \hat{\Omega}^k, W^k, \Lambda^k) - \Psi_{\tau, \beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k) \\ & \leq \max \left\{ 1 - \tau, \frac{(\tau - 1)\tau^2}{1 + \tau - \tau^2} \right\} \beta \left(\|W^k - G^{k+1}\|_F^2 - \|W^k - G^k\|_F^2 \right). \end{aligned} \quad (22)$$

Summing up the relations (19)–(22), we show that the inequality (13) holds. $\blacksquare$

Remark 3.3: If the penalty parameter β satisfies the lower bound condition

$$\beta \geq \beta_1 := \max \left\{ 2 \max \left\{ \frac{1}{\tau}, \frac{\tau^2}{1 + \tau - \tau^2} \right\} \frac{\theta_{\hat{\Omega}}^2}{\theta_{\hat{\mathcal{A}}}^2}, \theta_W \sqrt{2 \max \left\{ \frac{1}{\tau}, \frac{\tau^2}{1 + \tau - \tau^2} \right\}} \right\} \quad (23)$$

with $\theta_{\hat{\Omega}} := n \sqrt{\sum_{i=1}^n \sum_{j=1}^n \|x_i^e \hat{A}_j\|_\infty^2}$ and $\theta_W := \sqrt{n \|x^e\|_\infty^2 \|x^e\|_2^2}$. According to Lemma 3.1, the sequence $\{(G^k, \hat{\Omega}^k, W^k, \Lambda^k)\}_{k=1}^\infty$ is decreasing. Moreover, the expression β_1 in the right-hand side of (23) provides us a computable thresholding for the penalty parameter β , which is crucial to the corresponding numerical algorithm design, see Section 4 for more details.

We next show the boundedness of the sequence generated by Algorithm 1.

Lemma 3.2: Suppose that $\tau \in (0, (1 + \sqrt{5})/2)$ and Assumption 3.1 holds. If the following two conditions are satisfied:

- (The penalty parameter condition) $\beta > \beta_2 := \max\{\beta_1, n \|x^e\|_\infty^2 \max\{1/\tau, \tau\}\}$, where β_1 is defined as in (23);
- (The first iterative condition) $\Psi_{\tau, \beta}(G^1, \hat{\Omega}^1, W^1, \Lambda^1) < \liminf_{\|G\|_F \rightarrow +\infty} \Phi(G - G^e)$.

Then the sequence $\{(G^k, \hat{\Omega}^k, W^k, \Lambda^k)\}_{k=1}^\infty$ is bounded.

Proof: From the observation in Remark 3.3, we have

$$\Psi_{\tau, \beta}(G^1, \hat{\Omega}^1, W^1, \Lambda^1) \geq \Psi_{\tau, \beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k), \quad \forall k \geq 1. \quad (24)$$

The definition of $\Psi_{\tau,\beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k)$ given as in (12) implies that

$$\begin{aligned}
& \Psi_{\tau,\beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k) \\
&= l_\beta(G^k, \hat{\Omega}^k, W^k, \Lambda^k) + \max \left\{ 1 - \tau, \frac{(\tau - 1)\tau^2}{1 + \tau - \tau^2} \right\} \beta \|W^k - G^k\|_F^2 \\
&\geq \Phi(G^k - G^e) + \frac{1}{2} \|\hat{\mathcal{A}}^*(\hat{\Omega}^k) - W^k x^e - c^e\|_2^2 + \langle \Lambda^k, W^k - G^k \rangle \\
&\quad + \frac{\beta}{2} \|W^k - G^k\|_F^2 \\
&\quad + \max \left\{ 1 - \tau, \frac{(\tau - 1)\tau^2}{1 + \tau - \tau^2} \right\} \beta \|W^k - G^k\|_F^2 \\
&\geq \Phi(G^k - G^e) + \frac{1}{2} \|\hat{\mathcal{A}}^*(\hat{\Omega}^k) - W^k x^e - c^e\|_2^2 + \frac{\beta}{2} \left\| W^k - G^k + \frac{1}{\beta} \Lambda^k \right\|_F^2 \\
&\quad - \frac{1}{2\beta} \|\Lambda^k\|_F^2 + \max \left\{ 1 - \tau, \frac{(\tau - 1)\tau^2}{1 + \tau - \tau^2} \right\} \beta \|W^k - G^k\|_F^2. \tag{25}
\end{aligned}$$

From Step 4 of Algorithm 1 and Equation (11), we see that $\mathcal{L}_{x^e}(c^e + W^k x^e - \hat{\mathcal{A}}^*(\hat{\Omega}^k)) + \Lambda^{k-1} + \beta(W^k - G^k) = 0$ and $\Lambda^k = \Lambda^{k-1} + \tau\beta(W^k - G^k)$, which deduce that

$$- \tau \Lambda^k = \tau \mathcal{L}_{x^e}(c^e + W^k x^e - \hat{\mathcal{A}}^*(\hat{\Omega}^k)) + (1 - \tau)(\Lambda^k - \Lambda^{k-1}) \tag{26}$$

and the underlying two relations are valid, i.e.,

$$\begin{cases} \|\Lambda^k\|_F^2 \leq \frac{1}{\tau} \|\mathcal{L}_{x^e}(\hat{\mathcal{A}}^*(\hat{\Omega}^k) - c^e - W^k x^e)\|_F^2 + \frac{1-\tau}{\tau^2} \|\Lambda^k - \Lambda^{k-1}\|_F^2, & \text{if } \tau \in (0, 1], \\ \|\Lambda^k\|_F^2 \leq \tau \|\mathcal{L}_{x^e}(\hat{\mathcal{A}}^*(\hat{\Omega}^k) - c^e - W^k x^e)\|_F^2 + \left(1 - \frac{1}{\tau}\right) \|\Lambda^k - \Lambda^{k-1}\|_F^2, & \text{if } \tau \in \left(1, \frac{1+\sqrt{5}}{2}\right). \end{cases}$$

These inequalities show that

$$\begin{aligned}
& \|\Lambda^k\|_F^2 \\
&\leq \max\{1/\tau, \tau\} \|\mathcal{L}_{x^e}(\hat{\mathcal{A}}^*(\hat{\Omega}^k) - c^e - W^k x^e)\|_F^2 \\
&\quad + \max\{1 - \tau, (\tau - 1)\tau\} \beta^2 \|W^k - G^k\|_F^2 \\
&\leq n \|x^e\|_\infty^2 \max\{1/\tau, \tau\} \|\hat{\mathcal{A}}^*(\hat{\Omega}^k) - c^e - W^k x^e\|_2^2 \\
&\quad + \max\{1 - \tau, (\tau - 1)\tau\} \beta^2 \|W^k - G^k\|_F^2.
\end{aligned}$$

Using the relation (25) and the penalty parameter condition, we have

$$\begin{aligned}
& \Psi_{\tau,\beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k) \\
& \geq \frac{1}{2} \left(1 - \frac{n\|x^e\|_\infty^2 \max\{1/\tau, \tau\}}{\beta} \right) \|\hat{\mathcal{A}}^*(\hat{\Omega}^k) - W^k x^e - c^e\|_2^2 \\
& \quad + \frac{\beta}{2} \left\| W^k - G^k + \frac{1}{\beta} \Lambda^k \right\|_F^2 \\
& \quad + \left(\max \left\{ 1 - \tau, \frac{(\tau - 1)\tau^2}{1 + \tau - \tau^2} \right\} - \frac{1}{2} \max\{1 - \tau, (\tau - 1)\tau\} \right) \beta \|W^k - G^k\|_F^2 \\
& \quad + \Phi(G^k - G^e), \tag{27}
\end{aligned}$$

where the coefficient of the first term of the right-hand side is strictly larger than 0 and the corresponding third one satisfies the following condition

$$\begin{aligned}
& \max \left\{ 1 - \tau, \frac{(\tau - 1)\tau^2}{1 + \tau - \tau^2} \right\} - \frac{1}{2} \max\{1 - \tau, (\tau - 1)\tau\} \\
& = \begin{cases} 1 - \tau, & 0 < \tau < 1, \\ 0, & \tau = 1, \\ \frac{\tau(\tau - 1)(\tau^2 + \tau + 1)}{1 + \tau - \tau^2}, & 1 < \tau < \frac{1 + \sqrt{5}}{2}. \end{cases}
\end{aligned}$$

In light of the relations (24) and (27), the sequences $\{\|\hat{\mathcal{A}}^*(\hat{\Omega}^k) - W^k x^e - c^e\|_2\}_{k=1}^\infty$ and $\{\|W^k - G^k + (1/\beta)\Lambda^k\|_F\}_{k=1}^\infty$ are bounded. Moreover, if $\tau = 1$, then Equation (26) now reduces to $\Lambda^k = \mathcal{L}_{x^e}(\hat{\mathcal{A}}^*(\hat{\Omega}^k) - W^k x^e - c^e)$. This together with the above facts imply that $\{\Lambda^k\}_{k=1}^\infty$ is bounded. On the other hand, if $\tau \in (0, (1 + \sqrt{5})/2)$ and $\tau \neq 1$, then $\|\Lambda^k\|_F \leq \beta(\|W^k - G^k + 1/\beta\Lambda^k\|_F + \|W^k - G^k\|_F)$, which obtains the boundedness of $\{\Lambda^k\}_{k=1}^\infty$ as well.

We next prove that $\{(G^k, \hat{\Omega}^k, W^k)\}_{k=1}^\infty$ is also bounded. Combining Equation (24) with the first iterative condition, we see that $\{G^k\}_{k=1}^\infty$ is bounded. The relation $\|W^k\|_F \leq \|W^k - G^k + (1/\beta)\Lambda^k\|_F + \|G^k\|_F + (1/\beta)\|\Lambda^k\|_F$ shows that $\{W^k\}_{k=1}^\infty$ is a bounded sequence. Due to the boundedness of $\{\|\hat{\mathcal{A}}^*(\hat{\Omega}^k) - W^k x^e - c^e\|_2\}$, $\{\hat{\Omega}^k\}_{k=1}^\infty$ is also bounded. $\blacksquare$

After these preparations, we establish the convergence result of Algorithm 1.

Theorem 3.3: *Suppose that Assumption 3.1, the penalty parameter condition and the first iterative condition in Lemma 3.2 hold. For any $\tau \in (0, 1 + \sqrt{5}/2)$, we have*

$$\begin{aligned}
& \text{(i) } \lim_{k \rightarrow \infty} (\|G^{k+1} - G^k\|_F + \|\hat{\Omega}^{k+1} - \hat{\Omega}^k\|_F + \|W^{k+1} - W^k\|_F + \|\Lambda^{k+1} \\
& \quad - \Lambda^k\|_F) = 0.
\end{aligned}$$

- (ii) Any cluster point $(G^*, \hat{\Omega}^*)$ of the sequence $\{(G^k, \hat{\Omega}^k)\}_{k=1}^\infty$ generated by Algorithm 1 is a stationary point of Problem S-ISDQP(G, c), i.e., $(G^*, \hat{\Omega}^*) \in S(G, \hat{\Omega})$.

Proof: From Lemma 3.2, the sequence $\{(G^k, \hat{\Omega}^k, W^k, \Lambda^k)\}_{k=1}^\infty$ generated by Algorithm 1 is bounded. Hence, there exist at least one cluster point $(G^*, \hat{\Omega}^*, W^*, \Lambda^*)$ and a subsequence $\{(G^{k_i}, \hat{\Omega}^{k_i}, W^{k_i}, \Lambda^{k_i})\}_{i=1}^\infty$ such that $\lim_{i \rightarrow \infty} (G^{k_i}, \hat{\Omega}^{k_i}, W^{k_i}, \Lambda^{k_i}) = (G^*, \hat{\Omega}^*, W^*, \Lambda^*)$. Summing up these inequalities (13) over $k = 1, \dots, k_i - 1$, we have

$$\begin{aligned} & \Psi_{\tau, \beta}(G^1, \hat{\Omega}^1, W^1, \Lambda^1) - \Psi_{\tau, \beta}(G^{k_i}, \hat{\Omega}^{k_i}, W^{k_i}, \Lambda^{k_i}) \\ & \geq \left[\frac{\theta_{\hat{\Omega}}^2}{2} - \max \left\{ \frac{1}{\tau}, \frac{\tau^2}{1 + \tau - \tau^2} \right\} \frac{\theta_{\hat{\Omega}}^2}{\beta} \right] \sum_{k=1}^{k_i-1} \|\hat{\Omega}^{k+1} - \hat{\Omega}^k\|_F^2 \\ & \quad + \left[\frac{\beta}{2} - \max \left\{ \frac{1}{\tau}, \frac{\tau^2}{1 + \tau - \tau^2} \right\} \frac{\theta_W^2}{\beta} \right] \sum_{k=1}^{k_i-1} \|W^{k+1} - W^k\|_F^2. \end{aligned} \quad (28)$$

Taking the limit of both sides of (28), we obtain

$$\begin{aligned} & \Psi_{\tau, \beta}(G^1, \hat{\Omega}^1, W^1, \Lambda^1) - \Psi_{\tau, \beta}(G^*, \hat{\Omega}^*, W^*, \Lambda^*) \\ & \geq \left[\frac{\theta_{\hat{\Omega}}^2}{2} - \max \left\{ \frac{1}{\tau}, \frac{\tau^2}{1 + \tau - \tau^2} \right\} \frac{\theta_{\hat{\Omega}}^2}{\beta} \right] \sum_{k=1}^\infty \|\hat{\Omega}^{k+1} - \hat{\Omega}^k\|_F^2 \\ & \quad + \left[\frac{\beta}{2} - \max \left\{ \frac{1}{\tau}, \frac{\tau^2}{1 + \tau - \tau^2} \right\} \frac{\theta_W^2}{\beta} \right] \sum_{k=1}^\infty \|W^{k+1} - W^k\|_F^2. \end{aligned}$$

The above inequality and the penalty parameter condition imply that $\sum_{k=1}^\infty \|\hat{\Omega}^{k+1} - \hat{\Omega}^k\|_F^2 < +\infty$ and $\sum_{k=1}^\infty \|W^{k+1} - W^k\|_F^2 < +\infty$. Hence, $\hat{\Omega}^{k+1} - \hat{\Omega}^k \rightarrow 0$ and $W^{k+1} - W^k \rightarrow 0$. Similarly, summing up the inequality (18) over $k = 1, \dots, k_i - 1$ and taking the limit of both sides, we have

$$\begin{aligned} & \sum_{k=1}^\infty \|\Lambda^{k+1} - \Lambda^k\|_F^2 \\ & \leq \max \left\{ \frac{1}{\tau}, \frac{\tau^2}{1 + \tau - \tau^2} \right\} \\ & \quad \tau \left(\theta_{\hat{\Omega}}^2 \sum_{k=1}^\infty \|\hat{\Omega}^{k+1} - \hat{\Omega}^k\|_F^2 + \theta_W^2 \sum_{k=1}^\infty \|W^{k+1} - W^k\|_F^2 \right) \\ & \quad + \max \left\{ 1 - \tau, \frac{(\tau^2 - \tau)\tau}{1 + \tau - \tau^2} \right\} \tau \beta^2 \|W^1 - G^1\|_F^2 < +\infty, \end{aligned}$$

which means $\Lambda^{k+1} - \Lambda^k \rightarrow 0$. It follows from Step 4 of Algorithm 1 that $G^{k+1} - G^k = W^{k+1} - W^k - \frac{1}{\tau\beta}(\Lambda^{k+1} - \Lambda^k - (\Lambda^k - \Lambda^{k-1})) \rightarrow 0$, which shows statement (i).

We next prove statement (ii). Since $\Psi_{\tau,\beta}(G, \hat{\Omega}, W, \Lambda)$ is a proper, closed function, we have

$$\begin{aligned} & \liminf_{i \rightarrow +\infty} \Psi_{\tau,\beta}(G^{k_i+1}, \hat{\Omega}^{k_i+1}, W^{k_i+1}, \Lambda^{k_i+1}) \\ & \geq \Phi(G^* - G^e) + \frac{1}{2} \|\hat{\mathcal{A}}^*(\hat{\Omega}^*) - W^*x^e - c^e\|_2^2 + \langle \Lambda^*, W^* - G^* \rangle \\ & \quad + \frac{\beta}{2} \|W^* - G^*\|_F^2 \\ & \quad + \max \left\{ 1 - \tau, \frac{(\tau - 1)\tau^2}{1 + \tau - \tau^2} \right\} \|W^* - G^*\|_F^2. \end{aligned} \quad (29)$$

According to the fact that $\hat{\Omega}^{k_i+1}$ is an optimal solution of the $\hat{\Omega}$ -subproblem $\min_{\Omega \in S^p} l_\beta(G^{k_i+1}, \hat{\Omega}, W^{k_i}, \Lambda^{k_i})$ and the relation between l_β and $\Psi_{\tau,\beta}$, we conclude that $\Psi_{\tau,\beta}(G^{k_i+1}, \hat{\Omega}^{k_i+1}, W^{k_i}, \Lambda^{k_i}) \leq \Psi_{\tau,\beta}(G^{k_i+1}, \hat{\Omega}^*, W^{k_i}, \Lambda^{k_i})$ and the following relation

$$\begin{aligned} & \limsup_{i \rightarrow +\infty} \Psi_{\tau,\beta}(G^{k_i+1}, \hat{\Omega}^{k_i+1}, W^{k_i+1}, \Lambda^{k_i+1}) \\ & \leq \Phi(G^* - G^e) + \frac{1}{2} \|\hat{\mathcal{A}}^*(\hat{\Omega}^*) - W^*x^e - c^e\|_2^2 + \langle \Lambda^*, W^* - G^* \rangle \\ & \quad + \frac{\beta}{2} \|W^* - G^*\|_F^2 \\ & \quad + \max \left\{ 1 - \tau, \frac{(\tau - 1)\tau^2}{1 + \tau - \tau^2} \right\} \|W^* - G^*\|_F^2. \end{aligned} \quad (30)$$

In light of the relations (29) and (30), one can see that $\lim_{i \rightarrow \infty} \|\hat{\mathcal{A}}^*(\hat{\Omega}^{k_i+1}) - W^{k_i+1}x^e - c^e\|_2^2 = \|\hat{\mathcal{A}}^*(\hat{\Omega}^*) - W^*x^e - c^e\|_2^2$. This together with statement (i) yield that

$$\begin{cases} 0 \in \partial\Phi(G^* - G^e) + \mathcal{N}_{S_+^n}(G^*) - \Lambda^* + \beta(G^* - W^*), \\ 0 \in \hat{\mathcal{A}}(\hat{\mathcal{A}}^*(\hat{\Omega}^*) - W^*x^e - c^e) + \mathcal{N}_{S_+^p}(\hat{\Omega}^*), \\ \mathcal{L}_{x^e}(c^e + W^*x^e - \hat{\mathcal{A}}^*(\hat{\Omega}^*)) + \Lambda^* + \beta(W^* - G^*) = 0, \\ W^* - G^* = 0. \end{cases}$$

By using a simple calculation, the above system reduces to

$$\begin{cases} 0 \in \partial\Phi(G^* - G^e) + \mathcal{L}_{x^e}(c^e + G^*x^e - \hat{\mathcal{A}}^*(\hat{\Omega}^*)) + \mathcal{N}_{S_+^n}(G^*), \\ 0 \in \hat{\mathcal{A}}(\hat{\mathcal{A}}^*(\hat{\Omega}^*) - G^*x^e - c^e) + \mathcal{N}_{S_+^p}(\hat{\Omega}^*). \end{cases}$$

This shows that $(G^*, \hat{\Omega}^*)$ is a stationary point of Problem S-ISDQP(G, c), i.e., $(G^*, \hat{\Omega}^*) \in S(G, \hat{\Omega})$. ■

Assumption 3.2: The potential function $\Psi_{\tau,\beta}(G, \hat{\Omega}, W, \Lambda)$ is a KL function for any given $\tau \in (0, (1 + \sqrt{5})/2)$ and $\beta \geq \beta_2$, where β_2 is defined as in Lemma 3.2

Remark 3.4: This assumption was previously used in [18–21] to prove the convergence of the proximal alternating linearized minimization algorithm for nonconvex and nonsmooth problems, in which the KL property in the uniform sense as in Lemma 2.3 plays an crucial role.

With the above additional assumption on the potential function $\Psi_{\tau,\beta}(G, \hat{\Omega}, W, \Lambda)$, we next prove that the whole sequence generated by Algorithm 1 is convergent under the same assumptions in Theorem 3.3.

Theorem 3.4: Suppose that Assumptions 3.1 and 3.2, the penalty parameter condition and the first iterative condition in Lemma 3.2 hold. For any $\tau \in (0, (1 + \sqrt{5})/2)$, the whole sequence $\{(G^k, \hat{\Omega}^k)\}_{k=1}^\infty$ generated by Algorithm 1 converges to a stationary point of Problem S-ISDQP(G, c).

Proof: From the first iterative condition and Remark 3.3, the potential function $\Psi_{\tau,\beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k)$ is a decreasing sequence and bounded below. Hence, there exists a constant Ψ^* such that $\lim_{k \rightarrow \infty} \Psi_{\tau,\beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k) = \Psi^*$. Similar to the proof in Theorem 3.3, we only need to verify that $\{(G^k, \hat{\Omega}^k, W^k, \Lambda^k)\}_{k=1}^\infty$ is convergent. We discuss the following two cases:

- (A) There exists an index $\bar{k} \geq 1$ such that $\Psi_{\tau,\beta}(G^{\bar{k}}, \hat{\Omega}^{\bar{k}}, W^{\bar{k}}, \Lambda^{\bar{k}}) = \Psi^*$.
- (B) For any index $k \geq 1$, $\Psi_{\tau,\beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k) > \Psi^*$.

Case (A) From the fact that $\{\Psi_{\tau,\beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k)\}_{k=1}^\infty$ is a decreasing sequence, we have $\Psi_{\tau,\beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k) = \Psi^*$ for all $k \geq \bar{k}$. The inequality (13) in Lemma 3.1 implies that $\hat{\Omega}^k = \hat{\Omega}^{\bar{k}}$ and $W^k = W^{\bar{k}}$ for all $k \geq \bar{k}$, which show that $\{\hat{\Omega}^k\}_{k=1}^\infty$ and $\{W^k\}_{k=1}^\infty$ are convergent. Moreover, from Equation (15), we have $\|\Lambda^{k+1} - \Lambda^k\|_F = |1 - \tau| \|\Lambda^k - \Lambda^{k-1}\|_F = \dots = |1 - \tau|^{k+1-\bar{k}} \|\Lambda^{\bar{k}} - \Lambda^{\bar{k}-1}\|_F$ for all $k \geq \bar{k}$. Since $\tau \in (0, \frac{1+\sqrt{5}}{2})$, we obtain $0 < 1 - |1 - \tau| \leq 1$ and $\sum_{k=\bar{k}}^\infty \|\Lambda^{k+1} - \Lambda^k\|_F \leq 1/(1 - |1 - \tau|) \|\Lambda^{\bar{k}} - \Lambda^{\bar{k}-1}\|_F < +\infty$, which imply that $\{\Lambda^k\}_{k=1}^\infty$ is convergent. In view of the equation $G^{k+1} - G^k = W^{k+1} - W^k - (1/\tau\beta)(\Lambda^{k+1} - \Lambda^k - (\Lambda^k - \Lambda^{k-1}))$, we have $\|G^{k+1} - G^k\|_F \leq \|W^{k+1} - W^k\|_F + (1/\tau\beta)(\|\Lambda^{k+1} - \Lambda^k\|_F + \|\Lambda^k - \Lambda^{k-1}\|_F)$. This together with the convergence of $\{(W^k, \Lambda^k)\}_{k=1}^\infty$ show that $\{G^k\}_{k=1}^\infty$ is also convergent. Hence, $\{(G^k, \hat{\Omega}^k, W^k, \Lambda^k)\}_{k=1}^\infty$ is a convergent sequence under Case (A).

Case (B) Due to the boundedness of $\{(G^k, \hat{\Omega}^k, W^k, \Lambda^k)\}_{k=1}^\infty$, there exist at least one cluster point $(G^*, \hat{\Omega}^*, W^*, \Lambda^*)$. For simplicity, we denote by Ξ the set of cluster points of $\{(G^k, \hat{\Omega}^k, W^k, \Lambda^k)\}_{k=1}^\infty$. For any $(G^*, \hat{\Omega}^*, W^*, \Lambda^*) \in \Xi$, there exists a subsequence $\{(G^{k_i}, \hat{\Omega}^{k_i}, W^{k_i}, \Lambda^{k_i})\}_{i=1}^\infty$ such that $\lim_{i \rightarrow \infty} (G^{k_i}, \hat{\Omega}^{k_i}, W^{k_i}, \Lambda^{k_i}) =$

$(G^*, \hat{\Omega}^*, W^*, \Lambda^*)$. Similar to the proof of Theorem 3.3, from the lower semicontinuous property of $\Psi_{\tau, \beta}$, we obtain

$$\Psi^* = \lim_{i \rightarrow \infty} \Psi_{\tau, \beta}(G^{k_i}, \hat{\Omega}^{k_i}, W^{k_i}, \Lambda^{k_i}) \geq \Psi_{\tau, \beta}(G^*, \hat{\Omega}^*, W^*, \Lambda^*). \quad (31)$$

Since $\hat{\Omega}^{k_i}$ is an optimal solution of the $\hat{\Omega}$ -subproblem $\min_{\Omega \in \mathcal{S}^p} l_\beta(G^{k_i}, \hat{\Omega}, W^{k_i-1}, \Lambda^{k_i-1})$, we have

$$\begin{aligned} & \Psi_{\tau, \beta}(G^{k_i}, \hat{\Omega}^{k_i}, W^{k_i-1}, \Lambda^{k_i-1}) - \Psi_{\tau, \beta}(G^{k_i}, \hat{\Omega}^*, W^{k_i-1}, \Lambda^{k_i-1}) \\ &= l_\beta(G^{k_i}, \hat{\Omega}^{k_i}, W^{k_i-1}, \Lambda^{k_i-1}) - l_\beta(G^{k_i}, \hat{\Omega}^*, W^{k_i-1}, \Lambda^{k_i-1}) \leq 0. \end{aligned}$$

The above inequality together with Theorem 3.3 (i) yield that

$$\Psi^* = \lim_{i \rightarrow \infty} \Psi_{\tau, \beta}(G^{k_i}, \hat{\Omega}^{k_i}, W^{k_i}, \Lambda^{k_i}) \leq \Psi_{\tau, \beta}(G^*, \hat{\Omega}^*, W^*, \Lambda^*). \quad (32)$$

From (31) and (32), we see that $\Psi^* = \Psi_{\tau, \beta}(G^*, \hat{\Omega}^*, W^*, \Lambda^*)$ and $\Psi_{\tau, \beta}(G, \hat{\Omega}, W, \Lambda)$ is constant on the set Ξ . Since $\Psi_{\tau, \beta}(G, \hat{\Omega}, W, \Lambda)$ is a KL function, Lemma 2.3 implies that there exist $\epsilon > 0, \omega > 0$ and $\sigma \in \Gamma_\omega$ such that

$$\begin{aligned} & \sigma'(\Psi_{\tau, \beta}(G, \hat{\Omega}, W, \Lambda) - \Psi^*) \inf_{(Y_1, Y_2, Y_3, Y_4) \in \partial \Psi_{\tau, \beta}(G, \hat{\Omega}, W, \Lambda)} (\|Y_1\|_F + \|Y_2\|_F \\ &+ \|Y_3\|_F + \|Y_4\|_F) \geq 1, \end{aligned}$$

where the iterate $(G, \hat{\Omega}, W, \Lambda)$ satisfies the relations

$$\begin{cases} \inf_{(G^*, \hat{\Omega}^*, W^*, \Lambda^*) \in \Xi} (\|G - G^*\|_F + \|\hat{\Omega} - \hat{\Omega}^*\|_F + \|W - W^*\|_F + \|\Lambda - \Lambda^*\|_F) \leq \epsilon, \\ 0 < \Psi_{\tau, \beta}(G, \hat{\Omega}, W, \Lambda) - \Psi^* < \omega. \end{cases} \quad (33)$$

Since $\lim_{k \rightarrow \infty} (G^k, \hat{\Omega}^k, W^k, \Lambda^k) = (G^*, \hat{\Omega}^*, W^*, \Lambda^*)$ and $\lim_{k \rightarrow \infty} \Psi_{\tau, \beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k) = \Psi^*$, there exists an index k_1 such that the relation (33) holds at $(G^k, \hat{\Omega}^k, W^k, \Lambda^k)$ for all $k \geq k_1$ under the above parameters ϵ, ω . Consequently, we have

$$\begin{aligned} & \sigma'(\Psi_{\tau, \beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k) - \Psi^*) \inf_{(Y_1, Y_2, Y_3, Y_4) \in \partial \Psi_{\tau, \beta}^k} (\|Y_1\|_F + \|Y_2\|_F + \|Y_3\|_F + \|Y_4\|_F) \geq 1, \end{aligned} \quad (34)$$

where $\partial \Psi_{\tau,\beta}^k := \partial \Psi_{\tau,\beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k)$. We next calculate the subdifferential $\partial \Psi_{\tau,\beta}(G, \hat{\Omega}, W, \Lambda)$ at $(G^k, \hat{\Omega}^k, W^k, \Lambda^k)$. From the relation (12), we obtain

$$\begin{aligned}
 & \partial_G \Psi_{\tau,\beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k) \\
 &= \partial \Phi(G^k - G^e) + \mathcal{N}_{S_+^n}(G^k) - \Lambda^{k-1} + \beta(G^k - W^{k-1}) \\
 &+ \Lambda^{k-1} - \beta(G^k - W^{k-1}) - \Lambda^k \\
 &+ \left[\beta + 2\beta \max \left\{ 1 - \tau, \frac{(\tau-1)\tau^2}{1+\tau-\tau^2} \right\} \right] (G^k - W^k) \\
 &\ni \Lambda^{k-1} - \beta(G^k - W^{k-1}) - \Lambda^k \\
 &+ \left[\beta + 2\beta \max \left\{ 1 - \tau, \frac{(\tau-1)\tau^2}{1+\tau-\tau^2} \right\} \right] (G^k - W^k) \\
 &= \left[1 + 2 \max \left\{ \frac{1-\tau}{\tau}, \frac{(\tau-1)\tau}{1+\tau-\tau^2} \right\} \right] (\Lambda^{k-1} - \Lambda^k) + \beta(W^{k-1} - W^k),
 \end{aligned} \tag{35}$$

where the second relation uses Equation (9) and the last one is obtained by Step 4 of Algorithm 1. Similarly, it follows from Equation (10) and Equation (11) that

$$\begin{aligned}
 & \hat{\mathcal{A}}((W^{k-1} - W^k)x^e) \in \partial_{\hat{\Omega}} \Psi_{\tau,\beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k), \\
 & \partial_W \Psi_{\tau,\beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k) = \left[1 + 2 \max \left\{ \frac{1-\tau}{\tau}, \frac{(\tau-1)\tau}{1+\tau-\tau^2} \right\} \right] (\Lambda^k - \Lambda^{k-1}), \\
 & \partial_{\Lambda} \Psi_{\tau,\beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k) = W^k - G^k = \frac{1}{\tau\beta} (\Lambda^k - \Lambda^{k-1}).
 \end{aligned}$$

These relations show that there exists a positive constant ϖ such that

$$\begin{aligned}
 & \inf_{(Y_1, Y_2, Y_3, Y_4) \in \partial \Psi_{\tau,\beta}^k} (\|Y_1\|_F + \|Y_2\|_F + \|Y_3\|_F + \|Y_4\|_F) \\
 & \leq \varpi \left(\|W^k - W^{k-1}\|_F + \|\Lambda^k - \Lambda^{k-1}\|_F \right).
 \end{aligned} \tag{36}$$

For convenience, we denote

$$\begin{aligned}
 \Delta^k &:= \sigma(\Phi_{\tau,\beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k) - \Phi^*) - \sigma(\Phi_{\tau,\beta}(G^{k+1}, \hat{\Omega}^{k+1}, \\
 & W^{k+1}, \Lambda^{k+1}) - \Phi^*).
 \end{aligned}$$

From the decreasing property of $\{\Phi_{\tau,\beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k)\}$ and $\sigma \in \Gamma_\omega$, we obtain $\Delta^k \geq 0$, $\forall k \geq 1$ and the following relation holds for any $k \geq k_1$, that is,

$$\begin{aligned} & \varpi \left(\|W^k - W^{k-1}\|_F + \|\Lambda^k - \Lambda^{k-1}\|_F \right) \Delta^k \\ & \geq \inf_{(Y_1, Y_2, Y_3, Y_4) \in \partial \Psi_{\tau,\beta}^k} (\|Y_1\|_F + \|Y_2\|_F + \|Y_3\|_F + \|Y_4\|_F) \Delta^k \\ & \geq \Phi_{\tau,\beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k) - \Phi_{\tau,\beta}(G^{k+1}, \hat{\Omega}^{k+1}, W^{k+1}, \Lambda^{k+1}) \\ & \geq \frac{1}{2} \min\{b_1, b_2\} (\|\hat{\Omega}^{k+1} - \hat{\Omega}^k\|_F + \|W^{k+1} - W^k\|_F)^2, \end{aligned} \quad (37)$$

where the first inequality uses the relation (36) and $\sigma \in \Gamma_\omega$, the second one is obtained by (34) and the last one follows from (13), where b_1, b_2 are respectively given by

$$b_1 := \frac{\theta_{\hat{\mathcal{A}}}^2}{2} - \max \left\{ \frac{1}{\tau}, \frac{\tau^2}{1 + \tau - \tau^2} \right\} \frac{\theta_{\hat{\Omega}}^2}{\beta}, \quad b_2 := \frac{\beta}{2} - \max \left\{ \frac{1}{\tau}, \frac{\tau^2}{1 + \tau - \tau^2} \right\} \frac{\theta_W^2}{\beta}.$$

In view of (37), for any $\gamma > 0$, we have

$$\begin{aligned} & \frac{1}{2\gamma} \left(\|W^k - W^{k-1}\|_F + \|\Lambda^k - \Lambda^{k-1}\|_F \right) + \frac{\varpi \gamma \Delta^k}{\min\{b_1, b_2\}} \\ & \geq \|\hat{\Omega}^{k+1} - \hat{\Omega}^k\|_F + \|W^{k+1} - W^k\|_F. \end{aligned} \quad (38)$$

We conclude from Equation (15) and the inequalities (16)–(17) that

$$\begin{aligned} & \|\Lambda^k - \Lambda^{k-1}\|_F \\ & \leq \tau \left(\|\mathcal{L}_{x^e}(\hat{\mathcal{A}}^*(\hat{\Omega}^k - \hat{\Omega}^{k-1}))\|_F + \|\mathcal{L}_{x^e}((W^k - W^{k-1})x^e)\|_F \right) \\ & \quad + (1 - \tau) \|\Lambda^{k-1} - \Lambda^{k-2}\|_F \\ & \leq \tau (\theta_{\hat{\Omega}} \|\hat{\Omega}^k - \hat{\Omega}^{k-1}\|_F + \theta_W \|W^k - W^{k-1}\|_F) + |1 - \tau| \cdot \|\Lambda^{k-1} - \Lambda^{k-2}\|_F. \end{aligned} \quad (39)$$

Adding $-|1 - \tau| \cdot \|\Lambda^k - \Lambda^{k-1}\|_F$ to both sides of (39), we obtain

$$\begin{aligned} \|\Lambda^k - \Lambda^{k-1}\|_F & \leq d_1 \|\hat{\Omega}^k - \hat{\Omega}^{k-1}\|_F + d_2 \|W^k - W^{k-1}\|_F \\ & \quad + d_3 (\|\Lambda^{k-1} - \Lambda^{k-2}\|_F - \|\Lambda^k - \Lambda^{k-1}\|_F), \end{aligned}$$

where $d_1 := \tau/(1 - |1 - \tau|)\theta_{\hat{\Omega}}$, $d_2 := \tau/(1 - |1 - \tau|)\theta_W$ and $d_3 := \tau/(1 - |1 - \tau|)$. The above inequality and (38) imply that

$$\begin{aligned} & \left(1 - \frac{d_1}{2\gamma}\right) \|\hat{\Omega}^{k+1} - \hat{\Omega}^k\|_F + \left(1 - \frac{1+d_2}{2\gamma}\right) \|W^{k+1} - W^k\|_F \\ & \leq \frac{1+d_2}{2\gamma} \left(\|W^k - W^{k-1}\|_F - \|W^{k+1} - W^k\|_F\right) \\ & \quad + \frac{d_1}{2\gamma} \left(\|\hat{\Omega}^k - \hat{\Omega}^{k-1}\|_F - \|\hat{\Omega}^{k+1} - \hat{\Omega}^k\|_F\right) \\ & \quad + \frac{d_3}{2\gamma} \left(\|\Lambda^{k-1} - \Lambda^{k-2}\|_F - \|\Lambda^k - \Lambda^{k-1}\|_F\right) + \frac{\varpi\gamma\Delta^k}{\min\{b_1, b_2\}}. \end{aligned} \quad (40)$$

Summing up the above inequality (40) from $k = k_1$ to ∞ , we obtain

$$\begin{aligned} & \left(1 - \frac{d_1}{2\gamma}\right) \sum_{k=k_1}^{\infty} \|\hat{\Omega}^{k+1} - \hat{\Omega}^k\|_F + \left(1 - \frac{1+d_2}{2\gamma}\right) \sum_{k=k_1}^{\infty} \|W^{k+1} - W^k\|_F \\ & \leq \frac{1+d_2}{2\gamma} \|W^k - W^{k_1-1}\|_F + \frac{d_1}{2\gamma} \|\hat{\Omega}^k - \hat{\Omega}^{k_1-1}\|_F + \frac{d_3}{2\gamma} \|\Lambda^{k_1-1} - \Lambda^{k_1-2}\|_F \\ & \quad + \frac{\varpi\gamma}{\min\{b_1, b_2\}} \sigma(\Psi_{\tau, \beta}(G^{k_1}, \hat{\Omega}^{k_1}, W^{k_1}, \Lambda^{k_1}) - \Psi^*) < \infty. \end{aligned} \quad (41)$$

If we choose the parameter γ such that $1 - d_1/2\gamma > 0$ and $1 - (1 + d_2)/2\gamma > 0$, then $\sum_{k=k_1}^{\infty} \|\hat{\Omega}^{k+1} - \hat{\Omega}^k\|_F < \infty$, $\sum_{k=k_1}^{\infty} \|W^{k+1} - W^k\|_F < \infty$ and $\{\hat{\Omega}^k, W^k\}_{k=k_1}^{\infty}$ are convergent due to (41). Similarly, we have

$$\begin{aligned} & \sum_{k=k_1}^{\infty} \|\Lambda^k - \Lambda^{k-1}\|_F \leq \sum_{k=k_1}^{\infty} \left(d_1 \|\hat{\Omega}^k - \hat{\Omega}^{k-1}\|_F + d_2 \|W^k - W^{k-1}\|_F\right) \\ & \quad + d_3 \|\Lambda^{k_1-1} - \Lambda^{k_1-2}\|_F. \end{aligned}$$

This together with Step 4 of Algorithm 1 show that $\{\Lambda^k\}_{k=1}^{\infty}$ and $\{G^k\}_{k=1}^{\infty}$ are convergent. Hence, $\{(G^k, \hat{\Omega}^k, W^k, \Lambda^k)\}_{k=1}^{\infty}$ is also a convergent sequence under Case (B). ■

Notice that our convergence result relies on Assumption 3.1 and two conditions in Lemma 3.2, in which the first iterative condition seems to be strict for numerical algorithm designs. In order to deal with this issue, our target is to generate an initial point $(G^0, \hat{\Omega}^0, W^0, \Lambda^0)$ satisfying the following relations:

$$\Psi_{\tau, \beta}(G^1, \hat{\Omega}^1, W^1, \Lambda^1) \leq \Psi_{\tau, \beta}(G^0, \hat{\Omega}^0, W^0, \Lambda^0) < \liminf_{\|G\|_F \rightarrow +\infty} \Phi(G - G^e). \quad (42)$$

We next demonstrate that the left-hand side of (42) is valid under a suitable initial procedure.

Lemma 3.5: Suppose that Assumptions 3.1 and 3.2 hold and the penalty parameter β has the following condition

$$\beta \geq \beta_0 := \max \left\{ 4 \max \left\{ 1, \frac{\tau}{1 + \tau - \tau^2} \right\} \frac{\theta_{\hat{\Omega}}^2}{\theta_{\hat{\mathcal{A}}}^2}, 2\theta_W \sqrt{\max \left\{ 1, \frac{\tau}{1 + \tau - \tau^2} \right\}} \right\}.$$

For any given $\tau \in (0, (1 + \sqrt{5})/2)$, we choose the initial point $(G^0, \hat{\Omega}^0, W^0, \Lambda^0)$ such that $G^0 \in \mathcal{S}_+^n$, $W^0 \in \mathcal{S}^n$, $\hat{\Omega}^0 \in \mathcal{S}_+^p$ and $\Lambda^0 := \mathcal{L}_{x^e}(\hat{\mathcal{A}}^*(\hat{\Omega}^0) - W^0 x^e - c^e)$. Then, we have

$$\Psi_{\tau, \beta}(G^1, \hat{\Omega}^1, W^1, \Lambda^1) \leq \Psi_{\tau, \beta}(G^0, \hat{\Omega}^0, W^0, \Lambda^0). \quad (43)$$

Proof: From (11) and the definition of Λ^0 , we have

$$\begin{aligned} & W^1 - G^1 \\ &= \frac{1}{\beta} \left(\mathcal{L}_{x^e}(c^e + W^0 x^e - \hat{\mathcal{A}}^*(\hat{\Omega}^0)) - \mathcal{L}_{x^e}(c^e + W^1 x^e - \hat{\mathcal{A}}^*(\hat{\Omega}^1)) \right) \\ &= \frac{1}{\beta} \mathcal{L}_{x^e} \left((W^0 - W^1)x^e - \hat{\mathcal{A}}^*(\hat{\Omega}^0 - \hat{\Omega}^1) \right), \end{aligned} \quad (44)$$

Combining the relation (14) with Equation (44), we obtain

$$\begin{aligned} & \Psi_{\tau, \beta}(G^1, \hat{\Omega}^1, W^1, \Lambda^1) - \Psi_{\tau, \beta}(G^1, \hat{\Omega}^1, W^1, \Lambda^0) \\ &= \max \left\{ 1, \frac{\tau}{1 + \tau - \tau^2} \right\} \frac{1}{\beta} \left\| \mathcal{L}_{x^e} \left((W^0 - W^1)x^e - \hat{\mathcal{A}}^*(\hat{\Omega}^0 - \hat{\Omega}^1) \right) \right\|_F^2 \\ &\leq \max \left\{ 1, \frac{\tau}{1 + \tau - \tau^2} \right\} \frac{2}{\beta} \left(\theta_W^2 \|W^0 - W^1\|_F^2 + \theta_{\hat{\Omega}}^2 \|\hat{\Omega}^0 - \hat{\Omega}^1\|_F^2 \right) \\ &\quad - \max \left\{ 1 - \tau, \frac{(\tau - 1)\tau^2}{1 + \tau - \tau^2} \right\} \beta \|W^1 - G^1\|_F^2, \end{aligned} \quad (45)$$

where the last inequality uses the relations (16)–(17). Similar to the proof of Lemma 3.1, we also obtain the following relations:

$$\begin{aligned} & (a) \Psi_{\tau, \beta}(G^1, \hat{\Omega}^1, W^1, \Lambda^0) - \Psi_{\tau, \beta}(G^1, \hat{\Omega}^1, W^0, \Lambda^0) \\ &= \max \left\{ 1 - \tau, \frac{(\tau - 1)\tau^2}{1 + \tau - \tau^2} \right\} \beta (\|W^1 - G^1\|_F^2 - \|W^0 - G^1\|_F^2) \\ &\quad - \frac{\beta}{2} \|W^0 - W^1\|_F^2, \\ & (b) \Psi_{\tau, \beta}(G^1, \hat{\Omega}^1, W^0, \Lambda^0) - \Psi_{\tau, \beta}(G^1, \hat{\Omega}^0, W^0, \Lambda^0) \leq -\frac{\theta_{\hat{\mathcal{A}}}^2}{2} \|\hat{\Omega}^0 - \hat{\Omega}^1\|_F^2, \\ & (c) \Psi_{\tau, \beta}(G^1, \hat{\Omega}^0, W^0, \Lambda^0) - \Psi_{\tau, \beta}(G^0, \hat{\Omega}^0, W^0, \Lambda^0) \\ &\leq \max \left\{ 1 - \tau, \frac{(\tau - 1)\tau^2}{1 + \tau - \tau^2} \right\} \beta (\|W^0 - G^1\|_F^2 - \|W^0 - G^0\|_F^2). \end{aligned}$$

Summing up these relations with (45), we have

$$\begin{aligned}
& \Psi_{\tau,\beta}(G^1, \hat{\Omega}^1, W^1, \Lambda^1) - \Psi_{\tau,\beta}(G^0, \hat{\Omega}^0, W^0, \Lambda^0) \\
& \leq \left[\max \left\{ 1, \frac{\tau}{1+\tau-\tau^2} \right\} \frac{2\theta_W^2}{\beta} - \frac{\beta}{2} \right] \|W^0 - W^1\|_F^2 \\
& \quad + \left[\max \left\{ 1, \frac{\tau}{1+\tau-\tau^2} \right\} \frac{2\theta_{\hat{\Omega}}^2}{\beta} - \frac{\theta_{\hat{\mathcal{A}}}^2}{2} \right] \|\hat{\Omega}^0 - \hat{\Omega}^1\|_F^2 \\
& \quad - \max \left\{ 1 - \tau, \frac{(\tau-1)\tau^2}{1+\tau-\tau^2} \right\} \beta \|W^0 - G^0\|_F^2,
\end{aligned}$$

which shows that the relation (43) is true according to Remark 3.2 and $\beta \geq \beta_0$. $\blacksquare$

Remark 3.5: Consider the following initialization:

$$\begin{aligned}
G^0 &:= G^e + \kappa I_n, \quad W^0 := G^0, \quad \hat{\Omega}^0 := \Pi_{\mathcal{S}_+^p} \left(\frac{1}{\theta_{\hat{\mathcal{A}}}^2} \hat{\mathcal{A}}(W^0 x^e + c^e) \right), \\
\Lambda^0 &:= \mathcal{L}_{x^e}(\hat{\mathcal{A}}^*(\hat{\Omega}^0) - W^0 x^e - c^e).
\end{aligned}$$

where κ is a scaling parameter and $\kappa > \lambda_{\min}(G^e)$ (the minimal eigenvalue of G^e). One can easily observe that

$$\Psi_{\tau,\beta}(G^0, \hat{\Omega}^0, W^0, \Lambda^0) = \Phi(\kappa I_{\mathcal{S}_+^n}) + \frac{1}{2} \|\hat{\mathcal{A}}^*(\hat{\Omega}^0) - W^0 x^e - c^e\|_2^2. \quad (46)$$

In this paper, we choose $\phi(t)$ as one of the following functions:

- The l_1 regularizer: $\phi(t) := |t|$;
- The l_p regularizer: $\phi(t) := |t|^q$, $q \in (0, 1)$.

It follows from the coerciveness of these functions and Equation (46) that

$$\Psi_{\tau,\beta}(G^0, \hat{\Omega}^0, W^0, \Lambda^0) < \liminf_{\|G\|_F \rightarrow +\infty} \Phi(G - G^e),$$

which implies that the condition (42) holds for any given scaling parameter $\kappa > \lambda_{\min}(G^e)$ due to Lemma 3.5.

3.2. Non-ergodic iteration-complexity analysis

In this subsection, we present the non-ergodic iteration-complexity of obtaining an ϵ -approximate stationary point for Problem S-ISDQP(G, c).

We first introduce a technical lemma used in the following discussion, whose proof can be found in [29, Lemma 2.1].

Lemma 3.6: Let ξ_j be a nonnegative scalar sequence such that $\sum_{j=1}^{\infty} \xi_j < +\infty$. Then, we have $\min_{1 \leq j \leq k} \xi_j = o(1/k)$.

The next theorem presents that Algorithm 1 has an $o(1/k)$ non-ergodic iteration-complexity.

Theorem 3.7: Suppose that Assumptions 3.1 and 3.2, the penalty parameter condition and the first iterative condition in Lemma 3.2 hold. For any given $\tau \in (0, \frac{1+\sqrt{5}}{2})$, then

$$\min_{1 \leq j \leq k} \left\{ d^2 \left(0, \partial \Phi(G^{j+1} - G^e) + \mathcal{L}_{x^e}(c^e + G^{j+1}x^e - \hat{\mathcal{A}}^*(\hat{\Omega}^{j+1})) + \mathcal{N}_{S_+^n}(G^{j+1}) \right) + d^2 \left(0, \hat{\mathcal{A}}(\hat{\mathcal{A}}^*(\hat{\Omega}^{j+1}) - G^{j+1}x^e - c^e) + \mathcal{N}_{S_+^p}(\hat{\Omega}^{j+1}) \right) \right\} = o(1/k), \quad (47)$$

where $d(X, S)$ denotes the distance of a given matrix X to a set S under the Frobenius norm, i.e., $d(X, S) := \inf_{Y \in S} \|X - Y\|_F$.

Proof: According to the above assumptions and Theorem 3.4, the potential function $\Psi_{\tau, \beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k)$ is a decreasing sequence and there exists a constant Ψ^* such that $\lim_{k \rightarrow \infty} \Psi_{\tau, \beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k) = \Psi^*$. It follows from (13) and (18) that

$$\begin{aligned} & \Psi_{\tau, \beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k) - \Psi_{\tau, \beta}(G^{k+1}, \hat{\Omega}^{k+1}, W^{k+1}, \Lambda^{k+1}) \\ & \geq c_1(\|\hat{\Omega}^{k+1} - \hat{\Omega}^k\|_F^2 + \|W^{k+1} - W^k\|_F^2), \\ & \|\Lambda^{k+1} - \Lambda^k\|_F^2 \leq c_2(\|\hat{\Omega}^{k+1} - \hat{\Omega}^k\|_F^2 + \|W^{k+1} - W^k\|_F^2) \\ & + c_3(\|W^k - G^k\|_F^2 - \|W^{k+1} - G^{k+1}\|_F^2), \end{aligned}$$

where c_1, c_2, c_3 are respectively defined as

$$\begin{aligned} c_1 &:= \min \left\{ \left[\frac{\theta_{\hat{\mathcal{A}}}^2}{2} - \max \left\{ \frac{1}{\tau}, \frac{\tau^2}{1 + \tau - \tau^2} \right\} \frac{\theta_{\hat{\Omega}}^2}{\beta} \right], \right. \\ & \quad \left. \left[\frac{\beta}{2} - \max \left\{ \frac{1}{\tau}, \frac{\tau^2}{1 + \tau - \tau^2} \right\} \frac{\theta_W^2}{\beta} \right] \right\}, \\ c_2 &:= \max \left\{ 1, \frac{\tau^3}{1 + \tau - \tau^2} \right\} \cdot \max \left\{ \theta_{\hat{\Omega}}^2, \theta_W^2 \right\}, \\ c_3 &:= \max \left\{ 1 - \tau, \frac{(\tau^2 - \tau)\tau}{1 + \tau - \tau^2} \right\} \tau \beta. \end{aligned}$$

From these discussion, we have

$$\begin{aligned}
& \|\Lambda^{k+1} - \Lambda^k\|_F^2 + \|\hat{\Omega}^{k+1} - \hat{\Omega}^k\|_F^2 + \|W^{k+1} - W^k\|_F^2 \\
& \leq (1 + c_2) \left(\|\hat{\Omega}^{k+1} - \hat{\Omega}^k\|_F^2 + \|W^{k+1} - W^k\|_F^2 \right) \\
& \quad + c_3 (\|W^k - G^k\|_F^2 - \|W^{k+1} - G^{k+1}\|_F^2) \\
& \leq \frac{1 + c_2}{c_1} \left(\Psi_{\tau, \beta}(G^k, \hat{\Omega}^k, W^k, \Lambda^k) - \Psi_{\tau, \beta}(G^{k+1}, \hat{\Omega}^{k+1}, W^{k+1}, \Lambda^{k+1}) \right) \\
& \quad + c_3 (\|W^k - G^k\|_F^2 - \|W^{k+1} - G^{k+1}\|_F^2).
\end{aligned}$$

Summing the above inequality from $j=1$ to k , we conclude that

$$\begin{aligned}
\sum_{j=1}^k \xi_j & \leq \frac{1 + c_2}{c_1} \left(\Psi_{\tau, \beta}(G^1, \hat{\Omega}^1, W^1, \Lambda^1) - \Psi_{\tau, \beta}(G^{k+1}, \hat{\Omega}^{k+1}, W^{k+1}, \Lambda^{k+1}) \right) \\
& \quad + c_3 (\|W^1 - G^1\|_F^2 - \|W^{k+1} - G^{k+1}\|_F^2) \\
& \leq \frac{1 + c_2}{c_1} \left(\Psi_{\tau, \beta}(G^1, \hat{\Omega}^1, W^1, \Lambda^1) - \Psi^* \right) + c_3 \|W^1 - G^1\|_F^2 < +\infty,
\end{aligned} \tag{48}$$

where $\xi_j := \|\Lambda^{j+1} - \Lambda^j\|_F^2 + \|\hat{\Omega}^{j+1} - \hat{\Omega}^j\|_F^2 + \|W^{j+1} - W^j\|_F^2$ and the last inequality uses the first iterative condition in Lemma 3.2. Moreover, Lemma 3.6 and (48) show that

$$\min_{1 \leq j \leq k} \xi_j = \min_{1 \leq j \leq k} \left(\|\Lambda^{j+1} - \Lambda^j\|_F^2 + \|\hat{\Omega}^{j+1} - \hat{\Omega}^j\|_F^2 + \|W^{j+1} - W^j\|_F^2 \right) = o(1/k). \tag{49}$$

It follows from the relations (9)–(11) in Section 3.1 and Step 4 of Algorithm 1 that

$$\Lambda^k - \Lambda^{k+1} + \beta(W^k - G^{k+1}) \in \partial\Phi(G^{k+1} - G^e) + \mathcal{N}_{S_+^n}(G^{k+1}) - \Lambda^{k+1}, \tag{50}$$

$$\hat{\mathcal{A}}\left((W^k - W^{k+1})x^e\right) \in \hat{\mathcal{A}}(\hat{\mathcal{A}}^*(\hat{\Omega}^{k+1}) - W^{k+1}x^e - c^e) + \mathcal{N}_{S_+^p}(\hat{\Omega}^{k+1}), \tag{51}$$

$$\Lambda^{k+1} - \Lambda^k + \beta(G^{k+1} - W^{k+1}) = \mathcal{L}_{x^e}\left(c^e + W^{k+1}x^e - \hat{\mathcal{A}}^*(\hat{\Omega}^{k+1})\right) + \Lambda^{k+1}, \tag{52}$$

$$W^{k+1} = G^{k+1} + \frac{1}{\tau\beta}(\Lambda^{k+1} - \Lambda^k). \tag{53}$$

From Equations (50) and (52), we obtain $\beta(W^k - W^{k+1}) \in \partial\Phi(G^{k+1} - G^e) + \mathcal{L}_{x^e}(c^e + W^{k+1}x^e - \hat{\mathcal{A}}^*(\hat{\Omega}^{k+1})) + \mathcal{N}_{S_+^n}(G^{k+1})$. Substituting Equation (53) into

the right-hand sides of Equation (51) and the above inclusion, we have

$$\begin{aligned}
 & \beta(W^k - W^{k+1}) + \mathcal{L}_{x^e} \left(\frac{1}{\tau\beta} (\Lambda^k - \Lambda^{k+1}) x^e \right) \\
 & \in \partial\Phi(G^{k+1} - G^e) + \mathcal{L}_{x^e}(c^e + G^{k+1}x^e - \hat{A}^*(\hat{\Omega}^{k+1})) + \mathcal{N}_{S_+^n}(G^{k+1}), \\
 & \hat{A}((W^k - W^{k+1})x^e) + \hat{A} \left(\frac{1}{\tau\beta} (\Lambda^k - \Lambda^{k+1}) x^e \right) \\
 & \in \hat{A}(\hat{A}^*(\hat{\Omega}^{k+1}) - G^{k+1}x^e - c^e) + \mathcal{N}_{S_+^p}(\hat{\Omega}^{k+1}).
 \end{aligned}$$

Thus, we conclude from these inclusions that

$$\begin{aligned}
 & d^2 \left(0, \partial\Phi(G^{j+1} - G^e) + \mathcal{L}_{x^e}(c^e + G^{j+1}x^e - \hat{A}^*(\hat{\Omega}^{j+1})) + \mathcal{N}_{S_+^n}(G^{j+1}) \right) \\
 & + d^2 \left(0, \hat{A}(\hat{A}^*(\hat{\Omega}^{j+1}) - G^{j+1}x^e - c^e) + \mathcal{N}_{S_+^p}(\hat{\Omega}^{j+1}) \right) \\
 & \leq \left\| \beta(W^j - W^{j+1}) + \mathcal{L}_{x^e} \left(\frac{1}{\tau\beta} (\Lambda^j - \Lambda^{j+1}) x^e \right) \right\|_F^2 \\
 & + \left\| \hat{A}((W^j - W^{j+1})x^e) + \hat{A} \left(\frac{1}{\tau\beta} (\Lambda^j - \Lambda^{j+1}) x^e \right) \right\|_F^2.
 \end{aligned}$$

This together with Lemma 3.1 and Equation (49) imply that the relation (47) is valid. ■

4. Numerical experiments

In this section, we report some implementation issues and numerical experiments conducted for testing efficiency of Algorithm 1. All experiments are run in MATLAB R2017b on a 64-bit PC with an Intel (R) Core(TM) i7-6500U of 2.50 GHz CPU and 8.00GB of RAM equipped with Windows 7 operating system.

4.1. Implementation details

(1) Test problem

Example: Let G^e , c^e and x^e denote a random $n \times n$ symmetric matrix, a random $n \times 1$ vector and a given feasible solution to Problem SDQP(G, c), respectively. For convenience, we set all elements of x^e be 1. The matrices A_i ($i = 1, \dots, n$) are $m \times m$ symmetric matrices with entries in $[-1, 1]$. In our tests, we also set the rank of Z^e to be equal to $m/4$, $m/2$ and $3m/4$, respectively.

(2) The choice of ϕ

According to Remark 3.5, we choose $\phi(t) := |t|$ or $\phi(t) = |t|^q$, $q \in (0, 1)$. For simplicity, we set $q = 1/2$. For other choices of q , see the discussion in Case 2 of (4b) below. In order to obtain the convergence result, we need to verify that our potential function satisfies the KL inequality. In practice, this work is very

difficult and technical. However, due to the separable structure of our choices of ϕ , it has a similar structure as [22, Example 5.4(b)].

(3) *The choice of initial point and penalty parameter*

Our algorithm starts its iterations as the initialization given as in Remark 3.5. Moreover, in view of Remark 3.3, Lemma 3.2 and Lemma 3.5, we set $\beta = \max\{\beta_0, \beta_1, \beta_2\}$.

(4) *The discussions of subproblems*

(4a) The G -subproblem: From the definition of $l_\beta(G, \hat{\Omega}^k, W^k, \Lambda^k)$ and the choice of ϕ , we discuss the following two cases.

Case 1: If $\phi(t) = |t|$, the above G -subproblem becomes a convex problem, i.e.,

$$\begin{aligned} \min \quad & \sum_{i=1}^n \sum_{j=1}^n |G_{ij} - G_{ij}^e| + \langle \Lambda^k, W^k - G \rangle + \frac{\beta}{2} \|W^k - G\|_F^2 \\ \text{s.t.} \quad & G \in \mathcal{S}_+^n. \end{aligned}$$

Due to the separable property of the above objective function, we obtain

$$G^{k+1} := \Pi_{\mathcal{S}_+^n}(\tilde{G}^{k+1}),$$

where the (i, j) -th component of $\tilde{G}^{k+1}$ ($i, j \in \{1, 2, \dots, n\}$) is given by

$$\tilde{G}_{ij}^{k+1} := \begin{cases} W_{ij}^k + \frac{1}{\beta}(\Lambda_{ij}^k - 1) & \text{if } \beta W_{ij}^k + \Lambda_{ij}^k - \beta G_{ij}^e > 1, \\ W_{ij}^k + \frac{1}{\beta}(\Lambda_{ij}^k + 1) & \text{if } \beta W_{ij}^k + \Lambda_{ij}^k - \beta G_{ij}^e < -1, \\ G_{ij}^e & \text{if } \beta W_{ij}^k + \Lambda_{ij}^k - \beta G_{ij}^e \in [-1, 1]. \end{cases}$$

Case 2: If $\phi(t) = |t|^{1/2}$, the above G -subproblem now is a nonconvex problem due to the appearance of $|t|^{1/2}$, which has the following form

$$\begin{aligned} \min \quad & \sum_{i=1}^n \sum_{j=1}^n |G_{ij} - G_{ij}^e|^{1/2} + \langle \Lambda^k, W^k - G \rangle + \frac{\beta}{2} \|W^k - G\|_F^2 \\ \text{s.t.} \quad & G \in \mathcal{S}_+^n. \end{aligned}$$

Similar to Case 2, we obtain

$$G^{k+1} := \Pi_{\mathcal{S}_+^n}(\hat{G}^{k+1}),$$

where the (i, j) -th component of $\hat{G}^{k+1}$ ($i, j \in \{1, 2, \dots, n\}$) is given by

$$\hat{G}_{ij}^{k+1} := G_{ij}^e + \text{Prox}_{\frac{1}{\beta}}^\phi(W_{ij}^k + \frac{1}{\beta}\Lambda_{ij}^k - G_{ij}^e).$$

Here $\text{Prox}_{1/\beta}^\phi$ denotes the proximal point of ϕ with the parameter $1/\beta$ in one-dimensional case, whose explicit expression can be found in [30, Theorem 1]

or [31, Section 3]. For other choices of q , it can be approximated by a iterative reweighted minimization algorithm, see [32] for more details.

(4b) The $\hat{\Omega}$ -subproblem: From the definition of $l_\beta(G^{k+1}, \hat{\Omega}, W^k, \Lambda^k)$, this problem is equivalent to

$$\begin{aligned} \min \quad & \frac{1}{2} \|\hat{\mathcal{A}}^*(\hat{\Omega}) - W^k x^e - c^e\|_2^2 \\ \text{s.t.} \quad & \hat{\Omega} \in \mathcal{S}_+^p. \end{aligned} \quad (54)$$

In view of the proof in Theorem 3.1 and the relationship between the above objective function and $\Psi_{\tau, \beta}$, we only need to solve $\hat{\Omega}^{k+1}$ satisfying the following condition

$$\begin{aligned} & \Psi_{\tau, \beta}(G^{k+1}, \hat{\Omega}^{k+1}, W^k, \Lambda^k) - \Psi_{\tau, \beta}(G^{k+1}, \hat{\Omega}^k, W^k, \Lambda^k) \\ &= \frac{1}{2} \|\hat{\mathcal{A}}^*(\hat{\Omega}^{k+1}) - W^k x^e - c^e\|_2^2 - \frac{1}{2} \|\hat{\mathcal{A}}^*(\hat{\Omega}^k) - W^k x^e - c^e\|_2^2 \\ &\leq -\frac{\theta^2}{2} \|\hat{\Omega}^k - \hat{\Omega}^{k+1}\|_F^2. \end{aligned} \quad (55)$$

Moreover, the above $\hat{\Omega}$ -subproblem (54) has a same structure as [15, Problem (40)] and hence we can adapt its corresponding spectral projection gradient solver equipped with the condition (55) to deal with (54).

(4c) The W -subproblem: According to the generalized Fermat's rule [28, Theorem 10.1], we obtain

$$\mathcal{L}_{x^e}(c^e + W^{k+1} x^e - \hat{\mathcal{A}}^*(\hat{\Omega}^{k+1})) + \Lambda^k + \beta(W^{k+1} - G^{k+1}) = 0,$$

which has a closed-form solution

$$W^{k+1} = -\mathcal{L}_{(x^e(x^e)^T + \beta I_n)}^{-1} \left(\Lambda^k - \beta G^{k+1} + \mathcal{L}_{x^e}(c^e - \hat{\mathcal{A}}^*(\hat{\Omega}^{k+1})) \right),$$

where $\mathcal{L}_{(x^e(x^e)^T + \beta I_n)}^{-1}$ denotes the inverse operator of $\mathcal{L}_{(x^e(x^e)^T + \beta I_n)}$ given as in (6).

(5) *The stopping criterion*

We terminate Algorithm 1 when

$$\text{Res} := \frac{\|G^{k+1} - G^k\|_F + \|\hat{\Omega}^{k+1} - \hat{\Omega}^k\|_F + \|W^{k+1} - W^k\|_F + \|\Lambda^{k+1} - \Lambda^k\|_F}{\|G^k\|_F + \|\hat{\Omega}^k\|_F + \|W^k\|_F + \|\Lambda^k\|_F + 1} \leq \text{Tol}$$

for some $\text{Tol} > 0$. In the following experiment, we set $\text{Tol} = 10^{-3}$.

4.2. Numerical results

In all the tables below, 'r', 'it1', 'it2', 'it3', 't1', 't2', 't3' and 'res1', 'res2', 'res3' denote the rank of Z^e , the number of iterations, the total computing time in seconds, the residual at the final iterate under the parameter $\tau = 0.8, 1.0, 1.2$, respectively.

Table 1. Numerical results under the l_1 regularizer.

Size		Rank	Iteration			Time			Residual		
n	m	r	it1	it2	it3	t1	t2	t3	res1	res2	res3
100	100	25	43	45	60	0.43	0.43	0.57	9.75e−4	9.64e−4	9.59e−4
		50	47	47	64	0.44	0.45	0.60	9.52e−4	9.92e−4	9.61e−4
		75	50	50	68	0.47	0.47	0.65	9.72e−4	9.48e−4	9.63e−4
500	500	125	49	50	68	24.65	25.24	34.15	9.02e−4	9.65e−4	9.83e−4
		250	52	52	75	25.56	25.73	37.30	9.75e−4	9.34e−4	9.29e−4
		375	54	54	77	25.88	26.30	38.19	9.84e−4	9.67e−4	9.91e−4
1000	1000	250	53	53	75	259.52	261.33	374.81	9.54e−4	9.52e−4	9.98e−4
		500	55	55	79	268.69	267.88	388.50	9.78e−4	9.42e−4	9.78e−4
		750	57	57	82	276.24	278.55	403.28	9.73e−4	9.03e−4	9.50e−4

Table 2. Numerical results under the $l_{1/2}$ regularizer.

Size		Rank	Iteration			Time			Residual		
n	m	r	it1	it2	it3	t1	t2	t3	res1	res2	res3
100	100	25	55	58	68	0.73	0.82	0.91	9.26e−4	9.78e−4	9.97e−4
		50	57	66	76	0.76	0.87	1.01	9.18e−4	9.08e−4	9.40e−4
		75	63	68	85	0.83	0.95	1.15	9.90e−4	9.38e−4	8.96e−4
500	500	125	50	50	72	30.52	31.49	44.11	9.60e−4	9.54e−4	9.38e−4
		250	54	54	77	32.64	33.41	46.88	9.21e−4	9.17e−4	9.78e−4
		375	55	55	79	32.94	34.63	47.92	9.49e−4	9.06e−4	9.88e−4
1000	1000	250	53	54	76	284.15	288.45	416.41	9.59e−4	9.98e−4	9.98e−4
		500	56	56	80	296.49	297.92	435.33	9.40e−4	9.20e−4	9.65e−4
		750	58	58	83	308.28	309.10	455.86	9.73e−4	9.07e−4	9.60e−4

Tables 1 and 2 show that the total computing time and the residual of Algorithm 1 for solving sparse inverse semidefinite quadratic programming problems with different sizes $(n, m) = (100, 100), (500, 500), (1000, 1000)$ under the l_1 regularizer and the $l_{1/2}$ regularizer, respectively. From the above numerical results, our algorithm can solve the given test problem successfully. Moreover, Algorithm 1 with the parameter $\tau = 0.8$ has a better numerical performance compared with other choices in the sense that it takes less CPU time and obtains comparable residuals, regardless of the convex or nonconvex regularizer.

5. Concluding remarks

In this paper, we study a nonconvex alternating direction method of multiplier (ADMM) for minimizing a class of sparse inverse semidefinite quadratic programming problems. In particular, our algorithm adapts the classical ADMM with a general dual stepsize to handle the given problems. Under mild conditions, we show that any cluster point of the sequence generated by the proposed approach satisfies the corresponding stationary condition. Moreover, based on the famous Kurdyka–Łojasiewicz property of the corresponding potential function, we establish convergence results of the whole iterates sequence and derive an $o(1/k)$ non-ergodic iteration-complexity. Numerical results show that our

algorithm is suitable to solve the given sparse inverse semidefinite quadratic programming problems.

Disclosure statement

No potential conflict of interest was reported by the authors.

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